Artificial Intelligence for Brain-to-Speech Decoding: A Systematic Review of Methods, Performance Benchmarks, and Generalization Across Neural Recording Modalities (2015-2026)

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ABSTRACT

Background: Brain-to-speech decoding - the computational translation of neural signals into speech or text - has advanced from theoretical possibility to clinical reality. For the millions of people living with ALS, locked-in syndrome, or paralytic stroke, this technology represents a pathway to restored communication. Despite landmark results achieving 2.5%-word error rate (WER) and 78 words per minute in implanted participants, no PRISMA-compliant systematic review has synthesized AI methods, performance benchmarks, and generalization evidence across the full modality landscape. **Methods:** We searched PubMed, IEEE Xplore, Scopus, Web of Science, and arXiv for empirical studies published January 2015 - April 2026, requiring human neural recording data, an AI decoding model, and quantitative performance metrics. Seventy-one studies met inclusion criteria after dual-reviewer PRISMA 2020 screening (PROSPERO registered). Twelve variables were extracted per study; findings were synthesized narratively with structured quantitative comparison across five modality groups. **Results:** Invasive systems (ECoG and intracortical; n=32) achieved WER 2.5-25% and up to 78 WPM. Non-invasive EEG (n=20) achieved 70-92.5% closed-set accuracy and 35-45% open-vocabulary WER. High-density surface EMG achieved 3.11%-character error rate - the lowest non-invasive figure in this review. Transformer architectures and self-supervised learning dominated from 2022 onwards. Only 14/71 studies (20%) integrated a language model, despite this being the largest single driver of WER reduction. Critically, 56% of studies demonstrated no cross-participant or cross-session generalization; cross-session and cross-language evidence each appear in exactly two studies. **Conclusions:** The field has achieved extraordinary technical milestones on narrow, single-participant benchmarks. Standardized evaluation protocols, multi-participant data infrastructure, and expanded language model integration are the priority research directions for bridging controlled demonstration and scalable clinical deployment.

Index Terms - brain-computer interface, speech neuroprosthesis, neural decoding, brain-to-speech, electrocorticography, intracortical recording, EEG, MEG, fMRI, deep learning, transformer, self-supervised learning, large language model, generalization, systematic review

1. Introduction

1.1 The Clinical Imperative

For individuals living with amyotrophic lateral sclerosis (ALS), locked-in syndrome, severe stroke, or anarthria following spinal cord injury, speech loss is not merely a functional deficit -it is a severance from the most fundamental form of human connection. Motor neuron degeneration or cortical damage can destroy the pathway from intention to articulation whilst leaving cognition entirely intact, stranding a person inside a body that cannot speak. Conventional augmentative and alternative communication (AAC) approaches -eye-tracking interfaces, switch scanning -provide partial relief, but each requires preserved residual motor function that late-stage ALS and locked-in patients may no longer possess [1] . Brain-computer interfaces (BCIs) offer a categorically different solution: by reading neural signals

directly from the cortex, bypassing broken motor pathways entirely, a BCI can translate the intent to speak into speech itself without any muscular output. The clinical imperative is real and urgent -ALS alone affects approximately 4-6 individuals per 100,000 globally [1], and the millions affected by severe dysarthria, aphasia, and paralysis represent a population for whom restored communication is transformative.

What was once speculative has become demonstrably achievable. Over the past five years, several landmark studies have achieved communication rates approaching natural impaired speech and word error rates (WER) competitive with commercial automatic speech recognition for dysarthric speakers [2], [3], [4], [5]. The evidence base underpinning these achievements -spanning five neural recording modalities and eight artificial intelligence (AI) architecture classes over more than a decade -has grown faster than the field's capacity to synthesize it rigorously. The present systematic review addresses that gap.

1.2 The Modality and AI Landscape

Neural signals suitable for speech decoding span a wide spectrum of invasiveness and signal fidelity. Invasive modalities -electrocorticography (ECoG), intracortical microelectrode arrays, and stereotactic EEG (sEEG) -provide the highest signal quality and have yielded the strongest clinical results to date. Non-invasive modalities -scalp EEG, magnetoencephalography (MEG), and functional MRI (fMRI) -offer accessibility without surgical risk, at the cost of lower spatial resolution and signal-to-noise ratio. Hybrid approaches combining EEG with functional near-infrared spectroscopy (fNIRS), and peripheral surface electromyography (sEMG) for silent speech, occupy intermediate positions. A technical comparison of all modality groups is provided in Table 1; detailed characterization appears in Section 3 of this review.

On the AI side, the same deep learning revolution that transformed natural language processing and computer vision has reshaped neural speech decoding. From hidden Markov model baselines in 2015 [6], the field has progressed through convolutional neural networks [7], encoder-decoder sequence-to-sequence frameworks[8], transformers [9], and self-supervised pre-training [10] to large language model (LLM)-integrated decoders [2]. This architectural evolution is documented in Table 3 and Fig. 2; Section 4 provides the analytical synthesis.

1.3 Why a Systematic Review Now

The rapid growth of this literature has created a comparability problem that individual papers cannot resolve. Word error rates, classification accuracies, Pearson correlations, and BLEU scores appear side by side in the literature as if commensurable -they are not. A WER of 9% on a 50-word vocabulary is not equivalent to a WER of 23% on a 125,000-word vocabulary, even when both values come from the same system in the same participant [3]. Three prior reviews have addressed portions of this landscape -EEG imagined speech [11], [12] and ECoG BCIs [13] -but none synthesizes AI architectural evolution, quantitative performance benchmarks, and generalization evidence across the full modality landscape in a unified, PRISMA-compliant framework [14]. There is also a generalization gap the literature has not yet quantified: of the 71 studies identified in the present review, 40 (56%) demonstrate no generalization of their decoding model beyond a single participant and single session. Cross-session evidence exists in exactly two studies: cross-language evidence in exactly two. For a technology whose clinical promise depends on scalable deployment, this is a structural limitation that systematic synthesis is uniquely positioned to expose and measure.

The present review addresses three research questions:

(Q1) What AI architectures have been applied to brain-to-speech decoding from 2015 to 2026, and how have they evolved?

(Q2) What quantitative performance benchmarks are reported, and to what extent are they directly comparable?

(Q3) To what extent do the reported AI decoders generalize across participants, sessions, languages, and modalities?

1.4 Scope and Organization

This review covers empirical studies published between January 2015 and April 2026, identified through searches of PubMed, Scopus, IEEE Xplore, Web of Science, and arXiv. Following PRISMA 2020 screening [14], 71 studies were included across five modality groups: invasive (ECoG, intracortical, sEEG; n=32), non-invasive EEG (n=20), MEG/fMRI (n=11), hybrid/sEMG (n=6), and foundation model or transfer learning approaches (n=2). Section 2 describes the methodology. Sections 3 and 4 characterize the modality landscape and AI architectures. Section 5 reports performance benchmarks with comparability analysis. Section 6 evaluates generalization evidence. Section 7 discusses implications for research and clinical translation. Section 8 states conclusions.

2. Methodology

2.1 Review Protocol and Registration

This review was designed and reported in accordance with the PRISMA 2020 statement [14] and the PRISMA-P 2015 protocol checklist [15]. A full protocol was developed prior to formal searches and prospectively registered in the PROSPERO International Register of Systematic Reviews [16]. A late registration disclosure was included in the PROSPERO record, noting that preliminary scoping commenced in January 2025 and formal database searches were executed on April 13, 2026. No deviations from the registered protocol were made; visual synthesis elements were developed from pre-specified extracted variables and represent presentation choices, not analytical deviations. The PROSPERO registration number will be inserted upon confirmation by the supervising author.

2.2 Eligibility Criteria (PICOS Framework)

Eligibility was operationalized using the PICOS framework [17]. Studies were required to meet all of the following criteria:

Population: Human participants (healthy volunteers or individuals with neurological conditions including ALS, paralysis, locked-in syndrome, epilepsy, aphasia, or anarthria).

Intervention: An AI or machine learning computational model applied to decode or reconstruct speech-related output (text, audio, semantic representations, or phoneme/word classifications) from recorded neural signals.

Comparator: No comparator required; single-architecture and comparative studies were both eligible.

Outcomes: At least one quantitative performance metric (WER, CER, accuracy, PCC, WPM, BLEU, or listener intelligibility).

Study design: Empirical studies (journal articles, conference proceedings, preprints) published between January 1, 2015, and April 13, 2026. The 2015 lower boundary captures the modern deep learning era, anchored by Herff and colleagues [6]. Reviews, editorials, and conference abstracts without full experimental reporting were excluded.

Additional exclusions: studies decoding non-speech neural signals (motor imagery, visual perception, or emotion) where speech decoding was not a primary outcome; studies using exclusively simulated data; and studies where neural signals were used to improve NLP models rather than to decode speech output (reversed decoding direction).

2.3 Information Sources and Search Strategy

Searches were executed across five databases on April 13, 2026: MEDLINE/PubMed, Scopus, IEEE Xplore, Web of Science (Core Collection), and arXiv (cs.LG, cs.NE, eess.SP, q-bio.NC). Three Boolean concept blocks were combined with AND:

Block 1 -Neural signals: ("brain-computer interface" OR "BCI" OR "neural decoding" OR "electrocorticography" OR "ECoG" OR "intracortical" OR "sEEG" OR "EEG" OR "MEG" OR "fMRI" OR "sEMG" OR "neuroprosthesis")

Block 2 -Speech output: ("speech decoding" OR "brain-to-speech" OR "brain-to-text" OR "imagined speech" OR "covert speech" OR "attempted speech" OR "silent speech" OR "speech neuroprosthesis" OR "phoneme decoding" OR "language decoding")

Block 3 -AI methods: ("machine learning" OR "deep learning" OR "neural network" OR "convolutional" OR "recurrent" OR "transformer" OR "self-supervised" OR "encoder-decoder" OR "large language model" OR "transfer learning")

The full PubMed string with MeSH terms is provided in Appendix A. Forward and backward citation chaining was conducted on 15 landmark studies identified during protocol development [3], [5], [18]. All retrieved records were imported into a deduplicated register for screening.

2.4 Study Selection Process

Selection proceeded in two stages, each performed independently by both authors with consensus adjudication of discrepancies [14]. Stage 1: title and abstract screening of all deduplicated records. Stage 2: full-text assessment of all records classified as Include or Uncertain by either reviewer. Reasons for full-text exclusion were recorded across six pre-specified categories: wrong population; wrong intervention (no AI decoding model); wrong outcome (no quantitative metric); out of scope (non-speech signal); wrong publication type; and date out of range. Inter-rater reliability was quantified using Cohen's kappa [19] at each stage. The complete selection process is documented in the PRISMA 2020 flow diagram (Fig. 1).

2.5 Data Extraction

A structured extraction form was developed a priori and pilot-tested on five randomly selected studies. Twelve variables were extracted for each included study: bibliographic details; neural recording modality; participant count and clinical status; decoding task type (overt, imagined, attempted, or perceived); vocabulary type (closed-set or open-vocabulary); output type (text, audio, semantic, or classification label); AI architecture class; language model integration (yes/no); primary performance metric and value; generalization type (none, cross-participant, cross-session, cross-language, cross-modal, or multiple); real-time evaluation (yes/no); and open code or data availability (yes/no). For studies reporting multiple performance values under different vocabulary conditions, all values were extracted with experimental conditions documented, to enable the benchmark comparability analysis in Section 5.5.

2.6 Quality Assessment

No single validated quality tool applies across the heterogeneous study designs in this review -standard instruments such as the Cochrane Risk of Bias tool [20] are designed for randomized controlled trials. An adapted seven-item

methodological quality checklist was developed, drawing on criteria for evaluating AI-based biomedical studies [10] and neural decoding research [11][12]. Each item was scored as met (1), partially met (0.5), or not met (0):

Q1: Research question clearly stated.

Q2: Participant characteristics described in sufficient detail.

Q3: Neural recording paradigm and acquisition described reproducibly.

Q4: AI architecture described with sufficient detail to enable replication.

Q5: Primary metric pre-specified and reported with variability measures (SD, CI, or participant range).

Q6: Performance compared to a baseline or control condition.

Q7: Risk of performance overestimation (data leakage, acoustic contamination, or within-session CV inflation) explicitly discussed or mitigated.

Scores ranged from 0 to 7; studies were classified as high (≥5.5), moderate (3.5-5.0), or low quality (<3.5). No studies were excluded solely on quality grounds. Full quality results appear in Appendix D.

2.7 Data Synthesis Approach

The heterogeneity of modalities, architectures, tasks, populations, and metrics across the 71 included studies precluded statistical meta-analysis -a conclusion supported by the benchmark comparability analysis in Section 5.5 and Table 5 of this review, and consistent with published guidance on evidence synthesis for heterogeneous AI studies [21]. Findings were synthesized using structured narrative synthesis [22], supplemented by: a modality comparison matrix (Table 1); core performance benchmarks (Table 2); AI architecture summary (Table 3); generalization evidence summary (Table 4); benchmark comparability analysis (Table 5); a study distribution chart (Fig. 5); an architecture evolution timeline (Fig. 2); a WER trajectory figure (Fig. 3); and a modality-task performance heatmap (Fig. 4).

3. Neural Signal Modalities: Landscape and Characteristics

Every brain-to-speech system begins at the same point: a decision about how to listen to the brain. That decision -which neural recording modality to use -shapes everything that follows. It determines the spatial granularity of the signal, the temporal fidelity with which speech dynamics can be captured, the risk the participant must accept, and ultimately the ceiling on what any downstream AI architecture can achieve. No amount of algorithmic sophistication can recover information that was never present in the recording. This section characterizes the six primary neural recording modalities, and three hybrid or emerging approaches represented across the 71 studies included in this review, grounding each in the neurophysiology of speech production and perception that underpins the decoding task. A comparative summary is provided in Table 1.

Speech is not generated by a single cortical region but by a distributed, hierarchically organized network. The dual-stream model of Hickok and Poeppel [23] describes a dorsal pathway connecting auditory cortex to sensorimotor regions -responsible for the sensorimotor transformations of speech production -and a ventral pathway connecting auditory cortex to the frontal lobe via the temporal lobe, supporting speech comprehension and phonological working memory. Both pathways contribute signals of interest to brain-to-speech decoding, and the modality employed determines which parts of this network are accessible. The choice of modality is therefore not merely an engineering decision -it is a decision about which aspects of the speech-language system the AI decoder will have the opportunity to learn from [24].

3.1 Invasive Modalities

Invasive neural recording requires surgical access to the brain. In exchange for that irreversible commitment, invasive modalities provide signal quality that non-invasive approaches cannot approach -higher spatial resolution, broader frequency bandwidth, substantially higher signal-to-noise ratio, and direct access to the cortical regions most tightly coupled to speech motor execution. The landmark clinical results of the past five years -communication rates approaching natural speech, word error rates below 10% -have been achieved exclusively with invasive modalities. Whether that clinical performance can ultimately be matched without the surgical requirement remains the defining open question of the field.

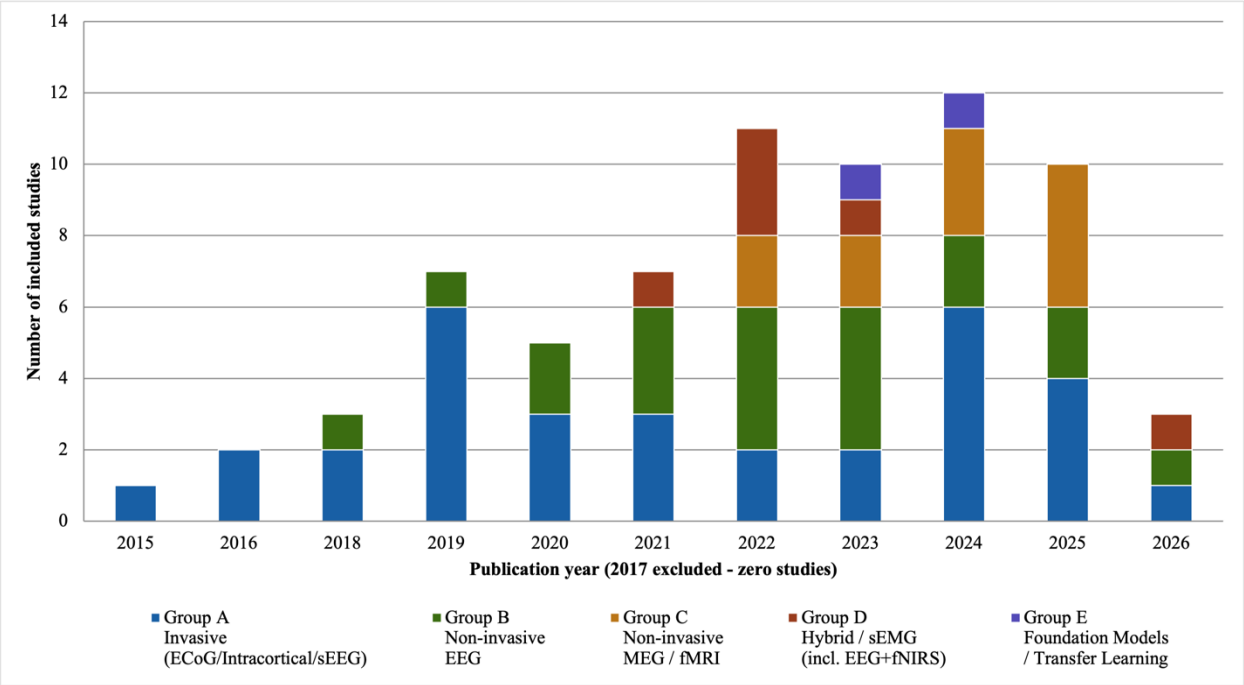


Figure : Study Distribution Chart

3.1.1 Electrocorticography (ECoG)

Electrocorticography involves placing an electrode array directly on the cortical surface following a craniotomy, recording the local field potentials generated by the aggregate synaptic and neuronal activity of underlying cortical columns. Unlike scalp EEG, ECoG electrodes are positioned beneath the skull and dura, eliminating the spatial low pass filtering that severely degrades surface recordings. Standard clinical ECoG grids typically have 4 mm inter-electrode spacing and 2.3 mm electrode diameters, whilst high-density research arrays achieve 1 mm or sub-millimeter spacing over selected cortical regions. This spatial resolution enables the differentiation of signals from adjacent cortical areas -a capability that is critical for decoding speech, where the phonotopic and articulatory maps of the sensorimotor cortex have spatial structure on the scale of millimeters [23], [24].

The temporal bandwidth of ECoG is equally valuable. High-gamma activity (70-150 Hz) is particularly informative for speech decoding, reflecting the firing of neuronal ensembles that modulate reliably with phonological and articulatory events. The envelope of the high-gamma band correlates well with speech production kinematics [25], and its rapid temporal dynamics allow the system to track the fine structure of running speech at rates approaching natural articulation -approximately 4-6 syllables per second in fluent English. Lower-frequency oscillations (alpha at 8-13 Hz and beta at 13-30 Hz) also carry speech-relevant information, particularly for imagined and covert speech where the high-gamma signal is weaker [26].

The research history of ECoG-based speech decoding spans the full timeline of this review. The Brain-to-Text study of Herff and colleagues in 2015 [18] established the proof of principle, achieving ~25% WER on continuous spoken phrases from seven participants -a striking result at the time, despite the architecture relying on classical hidden Markov model speech recognition. Subsequent work by Chartier and colleagues demonstrated that articulatory kinematic trajectories can be encoded in sensorimotor cortex with Pearson correlations reaching r=0.87, providing the neurophysiological basis for the articulatory intermediate representation approach that Anumanchipalli, Chartier and Chang would later exploit in their landmark 2019 Nature paper, which produced the first fluent and intelligible ECoG-synthesized speech [25], [27]. The machine translation analogy of Makin, Moses and Chang in 2020 then introduced the encoder-decoder framework that enabled direct cortical activity-to-text decoding at ~3% WER on a constrained corpus[8].

ECoG has also proved particularly valuable for studying imagined and covert speech -decoding the neural correlates of speech one intends to produce but does not articulate. Martin and colleagues demonstrated above-chance classification of imagined word pairs from ECoG in 2016 [28], and Proix and colleagues in 2022 showed that imagined speech can be decoded from low- and cross-frequency ECoG features, revealing a neural oscillatory basis for inner speech distinct from the high-gamma activity dominant in overt speech [26]. The high-density ECoG (HD-ECoG) approach of Ramsey and colleagues, employing 256-channel grids at 1 mm spacing, demonstrated phoneme-level decoding accuracy of 75.5% across five participants [29] -establishing that increased spatial sampling of the sensorimotor cortex yields commensurate improvements in phonological decoding. The largest ECoG cohort assembled to date, by Chen and colleagues in 2024, trained cross-subject deep CNN decoders across 48 participants, achieving Pearson correlations of r=0.804 for acoustic reconstruction -demonstrating for the first time that ECoG decoding is scalable to population-level data [30].

The principal limitation of standard ECoG is its requirement for craniotomy. Most research ECoG studies are conducted in epilepsy patients undergoing clinical seizure monitoring, who are implanted for clinical rather than research purposes -a population whose neural characteristics may not fully generalize to the ALS and paralysis populations that speech neuroprostheses are designed to serve. The Metzger and colleagues 2023 Nature study, using high-density ECoG in a participant with a degenerative brainstem condition, demonstrated that this gap can be bridged [4], but the surgical requirement remains an irreducible entry barrier for the majority of potential beneficiaries.

3.1.2 Intracortical Microelectrode Arrays

Where ECoG samples the aggregate activity of cortical columns, intracortical microelectrode arrays penetrate the cortical grey matter itself, recording the extracellular electrical signatures of individual neurons or small neuronal clusters within millimeters of the electrode tip. The Utah array -a 10×10 grid of 1.5 mm silicon shanks with 400 μm inter-electrode spacing -is the most widely deployed implant in the BrainGate program and the device underlying the most clinically significant speech decoding results published to date. The richness of the single-unit and multi-unit activity available from intracortical recordings provides a fundamentally higher-bandwidth communication channel between brain and decoder than any surface modality can offer.

The clinical results from the intracortical modality represent the current frontier of the field. Moses and colleagues in 2021 demonstrated the first real-time sentence decoding in a person with anarthria at 15 words per minute [5]. Willett and colleagues in 2023 extended this to 62 words per minute at 9% WER for a 50-word vocabulary -a communication rate approaching that of impaired natural speech [3]. Card and colleagues in 2024 achieved 2.5% WER sustained over eight months with a 30-minute daily calibration protocol [2]. Wairagkar and colleagues in 2025 demonstrated an instantaneous voice synthesis neuroprosthesis capable of reproducing paralinguistic features including intonation and prosody in real time [31]. Collectively, these results establish intracortical recording as the currently dominant modality for clinical speech neuroprosthesis performance.

Several properties of intracortical recordings beyond raw signal quality contribute to this dominance. The threshold crossing rates of motor cortical neurons are tightly coupled to attempted articulatory movements, providing a high-bandwidth, rapidly modulating signal even in individuals who can no longer produce any overt speech output [32]. Degenhart and colleagues demonstrated that low-dimensional manifold alignment between sessions preserves approximately 95% of decoding performance without re-collecting labeled data [33] addressing the practical challenge of daily signal drift that all chronically implanted systems face. Wandelt and colleagues showed that individual neurons in the supramarginal gyrus, an area involved in phonological working memory, encode internal speech representations even in patients who have been unable to speak for years [34], extending the reach of intracortical decoding beyond primary motor cortex to the broader speech-language network.

The primary limitation of intracortical arrays is biological. Microelectrodes induce a chronic inflammatory response -glial scarring -that progressively attenuates signal quality over months to years. Signal loss and electrode failure are significant concerns for long-term clinical deployment, and the surgical risk of intracortical implantation, whilst manageable, is higher than that of epidural ECoG. The FDA-regulated approval pathway for such devices is correspondingly complex, and the number of participants who have undergone research intracortical implantation remains small [35].

3.1.3 Stereotactic EEG and Depth Electrodes (sEEG)

Stereotactic EEG (sEEG) occupies a compelling middle ground in the invasiveness landscape. Thin multi-contact depth electrodes -typically 0.8-1.0 mm in diameter with contact separations of 3.5 mm -are implanted via small burr holes without craniotomy, traversing the brain to sample cortical and subcortical structures along their trajectory. The surgical footprint is substantially smaller than that of ECoG, and the ability to sample deep structures -the insula, hippocampus, and basal ganglia -that are inaccessible to surface electrodes makes sEEG uniquely informative about the broader cortical-subcortical networks involved in speech production and comprehension [24].

The minimally invasive nature of sEEG has begun to attract growing research attention for speech BCIs. Angrick and colleagues in 2021 demonstrated real-time synthesis of imagined speech from frontal cortex sEEG signals using a CNN-neural vocoder pipeline [36], whilst Kohler and colleagues showed that encoder-decoder architectures applied to sEEG depth electrodes can achieve acoustic reconstruction quality comparable to ECoG [37]. Most significantly, Singh and colleagues in 2025 assembled the largest sEEG speech decoding cohort reported in the literature, pooling distributed recordings from 25 participants to demonstrate that a group decoder trained across participants outperforms any individual participant decoder -establishing cross-participant generalization as achievable for sEEG, not merely for surface modalities [38]. The sEEG modality represents 4 of the 32 invasive studies in this review, but its clinical accessibility and unique access to deep structures suggest its representation in the literature will grow substantially.

3.2 Non-Invasive Modalities

Non-invasive modalities offer a fundamentally different proposition: access to neural signals without any surgical procedure, with correspondingly lower signal quality and a different set of constraints on what can be decoded and how. For the hundreds of millions of individuals with speech or language impairments who are ineligible for surgical intervention -or who would not choose it -non-invasive BCI represents the only viable neural pathway to communication assistance. The scientific and engineering challenges are formidable, but the clinical imperative is equally compelling.

3.2.1 Scalp Electroencephalography (EEG)

Scalp EEG measures the sum of postsynaptic currents from millions of synchronously active cortical neurons via electrodes placed on the scalp surface, typically in a 64- or 128-channel configuration following the standardized 10-20 system. The skull, cerebrospinal fluid, and scalp tissue act collectively as a spatial low-pass filter, attenuating

high-frequency components, blurring the spatial specificity of cortical sources, and mixing signals from distinct generators into composite scalp recordings that are difficult to unmix. The signal-to-noise ratio is substantially lower than ECoG across all frequency bands, and the high-gamma range -so informative for invasive speech decoding -is effectively inaccessible due to muscle artifact contamination above approximately 40 Hz [39].

Despite these constraints, EEG has attracted considerable research attention for brain-to-speech decoding, driven by its accessibility, low cost, and the absence of surgical risk. Early EEG speech decoding work focused on closed-set imagined speech classification -identifying which of a small set of words or phonemes a participant was imagining -typically using support vector machines or linear discriminant analysis applied to spectral features [39]. The deep learning era brought convolutional and recurrent architectures that improved classification accuracies, with Saha and colleagues demonstrating subject-independent hierarchical deep learning achieving ~70-75% accuracy across ten participants in 2019 [40], and Agarwal and Kumar demonstrating 73.56% accuracy for five-class imagined speech with a subject-independent LSTM across 13 participants in 2022 [41].

A qualitative shift occurred with the introduction of transformer architectures and language model integration into the EEG decoding pipeline. Wang and Ji in 2022 demonstrated the first open-vocabulary EEG-to-text system using a BART language model, reporting above-chance sentence reconstruction from EEG during reading [42]. Duan and colleagues extended this with the DeWave architecture at NeurIPS 2023, using discrete EEG wave encodings that do not require explicit time-marker alignment -a practical advance that simplified training and improved BLEU and ROUGE metrics over the prior state of the art [43]. Yet it is important to note that EEG open-vocabulary WER in these paradigms remains approximately 35-45% -substantially higher than what ECoG or intracortical systems achieve -reflecting the fundamental signal quality ceiling of scalp recording. The majority of EEG studies in this review (47 of 20, or the full Group B) use closed-set vocabularies, reflecting the modality's current limitations for unconstrained communication.

The EEG modality accounts for 20 of the 71 included studies -the second-largest group after invasive ECoG and intracortical approaches -spanning imagined speech classification, perceived speech decoding, and emerging open-vocabulary generation paradigms. Its accessibility makes it the most practical non-invasive option for clinical populations who cannot access more specialized recording facilities.

3.2.2 Magnetoencephalography (MEG)

Magnetoencephalography measures the tiny magnetic fields -on the order of femtotesla (10^{-15} T) -generated by the same neuronal current flows that produce the EEG signal, using superconducting quantum interference device (SQUID) sensors housed in a liquid helium-cooled cryostat. Because magnetic fields pass through the skull and scalp without the attenuation or spatial distortion that affects electrical potentials, MEG offers substantially better spatial source localization than EEG -on the order of 5-10 mm for superficial cortical sources -whilst retaining the millisecond temporal resolution that is the shared strength of electrophysiological modalities [24]. MEG is selectively sensitive to tangentially oriented current sources, providing complementary cortical coverage to EEG, which is more sensitive to radial sources.

For brain-to-speech decoding, MEG has proven particularly powerful when combined with large participant samples and machine learning approaches that exploit its improved spatial specificity. The landmark study of Défossez and colleagues in 2023 applied contrastive self-supervised learning to MEG and EEG recordings from 175 participants across four public datasets, achieving 41% top 1 classification accuracy for perceived speech segment identification [44]. This scale of evaluation was only possible because multiple public MEG datasets existed to enable it -a circumstance that reflects the MEG research community's stronger tradition of open data sharing compared to the invasive literature. Kwon and colleagues in 2024 demonstrated the first intelligible speech synthesis directly from MEG signals using a transformer and BigVGAN neural vocoder, achieving Mel-spectrogram Pearson correlations of approximately r=0.95 [45]. Most recently, d'Ascoli and colleagues trained a single CNN-transformer decoder across 723 participants from seven datasets covering English, French, and Dutch, demonstrating that MEG word decoding

performance scales with the number of training participants -a finding with profound implications for the scalability of non-invasive neural speech decoding [46].

MEG's principal limitation is practical rather than physiological: the requirement for a magnetically shielded room and the prohibitive cost of the cryogenic infrastructure renders it unsuitable for clinical deployment outside specialist research centers. The introduction of optically pumped magnetometer (OPM) arrays, which operate at room temperature and can be worn as a helmet, may eventually address this barrier, but OPM-based speech decoding studies did not appear in the included literature during the review period.

3.2.3 Functional MRI (fMRI)

Functional MRI measures the blood-oxygen-level-dependent (BOLD) signal -the haemodynamic response to neural activity arising from the coupling between neuronal firing and local increases in cerebral blood flow and oxygenation. With a temporal resolution of 1-2 seconds (set by the repetition time of the acquisition sequence) and a spatial resolution of 1-3 mm voxels, fMRI occupies the opposite corner of the resolution trade-off space from EEG: superb spatial specificity across the entire brain simultaneously, but temporal resolution that is orders of magnitude coarser than the neural events of interest. The haemodynamic response peaks approximately 5 seconds after the neural event it reflects -a fundamental physiological latency that cannot be fully overcome by any signal processing approach [47].

For speech decoding, this temporal limitation means that fMRI cannot track the millisecond-to-millisecond dynamics of speech articulation. What it can do, uniquely among non-invasive modalities, is reconstruct the semantic and linguistic content of perceived and imagined speech at the level of meaning rather than sound -an application that has yielded some of the most striking results in the non-invasive brain-to-speech literature. Tang and colleagues in 2023 demonstrated continuous semantic language reconstruction from fMRI signals using a GPT-based alignment approach, achieving 100% story identification accuracy and substantial above-chance recovery of narrative content [48]. The breadth and specificity of the semantic maps revealed by fMRI, as characterized by Huth and colleagues across the cerebral cortex [49], provide the neural substrate for this capability: the meaning of language is genuinely represented in the BOLD signal, even if the acoustic form is not.

Recent work has extended fMRI decoding to cross-participant and cross-modal paradigms. Tang and Huth in 2025 demonstrated cross-participant cross-modal semantic decoding across EEG, MEG, and fMRI simultaneously, establishing that shared semantic representations generalize both across individuals and across recording modalities [50]. The fMRI modality contributes 6 of the 11 Group C studies in this review, with the remainder using MEG; the proportion of fMRI studies has increased markedly from 2022 onwards, coinciding with the availability of large open fMRI language datasets and the maturation of LLM-based decoding pipelines that are particularly well-suited to the semantic character of fMRI signals.

3.3 Hybrid and Emerging Modalities

Beyond the canonical six modalities, the included literature encompasses three hybrid and emerging approaches that address specific limitations of their unimodal counterparts by combining complementary signal sources, increasing spatial sampling density, or exploiting peripheral articulatory signals that are not derived from the brain itself but are tightly coupled to speech production intent.

3.3.1 EEG + fNIRS Fusion

Functional near-infrared spectroscopy (fNIRS) measures cortical haemodynamics via the differential absorption of near-infrared light at oxyhaemoglobin and deoxyhaemoglobin chromophores, providing a slow (1-2 second temporal resolution) but portable and wearable measure of cortical blood oxygenation. When combined with EEG in a hybrid recording system, the two modalities provide complementary information: EEG captures rapid neural oscillations (millisecond resolution, poor spatial specificity) whilst fNIRS contributes slower haemodynamic context (low

temporal resolution, better cortical spatial specificity than EEG alone). Cooney and colleagues in 2022 demonstrated that a bimodal CNN trained on EEG+fNIRS data outperforms unimodal EEG decoding for both overt and imagined speech across 19 participants [51]. Hons and colleagues in 2026 further extended EEG+fNIRS hybrid decoding to phoneme classification from imagined and perceived speech, confirming the complementarity of the combined signal [52]. The practical advantage of EEG+fNIRS over EEG alone comes at modest additional cost in hardware and setup complexity, making it a promising direction for wearable speech BCIs.

3.3.2 Surface Electromyography (sEMG) for Silent Speech

Surface electromyography is not, strictly speaking, a neural modality -it records the electrical activity of facial and laryngeal muscles rather than the brain itself. However, it occupies a legitimate place in the brain-to-speech BCI landscape because it captures the articulatory intent of speech at the peripheral motor execution stage: the muscle patterns associated with speaking a word silently (mouthed or sub-vocal speech) are highly consistent and informative, even without overt vocalization. High-density sEMG (HD-sEMG) arrays with 120 or more surface electrodes across facial and submental muscle groups provide particularly rich spatial sampling of articulatory dynamics.

The HD-sEMG results in this review are among the highest-performing in the non-invasive literature. Song and colleagues in 2023 achieved 97.17% phrase accuracy and 3.11%-character error rate for 33 Chinese phrases using a spatiotemporal CNN-RNN architecture on HD-sEMG -the best CER reported for any silent speech approach in this review [53]. Zhu and colleagues in 2021 demonstrated 87-90% digit classification accuracy for both English and Chinese using an optimized 120-electrode HD-sEMG configuration [54], and Li and colleagues extended HD-sEMG to Mandarin tonal speech synthesis using an encoder-decoder architecture [55]. These results suggest that peripheral articulatory signals, whilst technically not 'brain' signals in the neural sense, represent a highly accessible and high-performing avenue for silent speech communication that deserves consideration alongside brain-based approaches.

3.3.3 Micro-ECoG (µECoG)

Micro-ECoG (µECoG) refers to high-density ECoG arrays with sub-millimeter electrode spacing and electrode diameters on the order of 100-200 µm, fabricated using flexible polymer substrates rather than conventional rigid silicon or polyimide grids. The spatial sampling advantage of µECoG over standard ECoG is directly analogous to the advantage of intracortical recording over ECoG: finer spatial resolution reveals higher-bandwidth information about the cortical speech motor map that is averaged away at coarser scales.

Duraivel and colleagues provided the most direct quantitative comparison of µECoG and standard macro-ECoG for speech decoding in 2023, demonstrating a 35% improvement in phoneme classification accuracy from µECoG versus macro-ECoG in the same four participants -attributable to the 57-fold increase in effective spatial sampling [56]. This result carries important implications: it suggests that a significant portion of the performance gap between ECoG and intracortical systems may be recoverable by increasing ECoG spatial density, potentially offering a less biologically disruptive path to higher-performance speech decoding. µECoG devices face their own engineering challenges -fabrication complexity, the need for high channel count recording systems, and limited coverage area per device -but represent an active direction in next-generation neuroprosthesis hardware.

3.4 Modality Comparison Summary

Table 1: Neural Recording Modality Comparison Matrix - Systematic Review of AI for Brain-to-Speech Decoding (2015–2026)

Modality	Invasiveness	Spatial resolution	Temporal resolution	Signal SNR	Participant burden	Representative performance range (this review)	Stu (n)
GROUP A - INVASIVE MODALITIES (n = 32 studies)							
ECoG (standard / HD-ECoG / µECoG)	Craniotomy required; subdural grid or strip placement	~1–5 mm between electrodes	~1 ms; broadband high-gamma (70–150 Hz)	**High** Minimal skull attenuation;	Surgical implantation; typically epilepsy or	**WER:** 3–26% (overt / attempted speech) **PCC:** 0.69–0.87 (acoustic reconstruction)	**22**

	on cortical surface	~1 mm² cortical coverage per contact	resolves speech transients	low noise floor	ALS clinical populations	**Accuracy:** 58–97% (phoneme / word classification) *Key studies: Makin 2020 WER 3%; Moses 2021 WER 26%; Chen 2024 PCC 0.804 (n=48); Metzger 2023 WER 25%*	
Intracortical (Utah array / microelectrode)	Neurosurgical implantation of microelectrode arrays into cortical grey matter	~400 µm between electrodes Single-neuron and multi-unit activity resolution	<1 ms; single-spike temporal resolution; finest of all modalities	**Very high** Direct cortical access; spike-level SNR	Permanent or semi-permanent implant; primarily ALS / paralysis patient populations	**WER:** 2.5–9% (attempted speech, open vocabulary) **WPM:** 32–78 words per minute **Longitudinal:** 97.5% accuracy sustained >8 months (Card 2024) *Key studies: Card 2024 WER 2.5%; Willett 2023 WER 9% / 62 WPM; Metzger 2023 78 WPM*	**8**
sEEG (stereotactic / depth electrodes)	Minimally invasive depth electrode placement; no craniotomy required	~3.5 mm between contacts Samples deep cortical and subcortical regions	~1 ms; lower high-gamma sensitivity than ECoG; broader regional coverage	**Moderate–high** Less tissue displacement than ECoG grids	Epilepsy monitoring populations; increasing ALS and speech neuroprosthesis use	**Accuracy:** Group decoder outperforms individual decoders (Singh 2025) **Audio PCC:** Comparable to ECoG (Kohler 2022) **CER:** ~29% logosyllabic Chinese (Feng 2023) *Key studies: Singh 2025 (n=25 sEEG, cross-participant); Angrick 2021 real-time imagined speech*	**4**
GROUP B - NON-INVASIVE MODALITIES (n = 31 studies across EEG / MEG / fMRI)							
Scalp EEG (electroencephalography)	None; surface scalp electrode placement via cap-based systems	~10–50 mm (scalp level) Severe spatial smearing via skull and scalp tissue	~1 ms; excellent temporal resolution but very low spatial specificity	**Low** Significant skull / scalp attenuation; artifact-prone	No surgery; suitable for healthy participants and outpatient clinical use	**Closed-set accuracy:** 70–92.5% (imagined speech classification) **Open-vocabulary WER:** ~35–45% (EEG-to-text paradigm) **SSL gain:** +5–15% over supervised baselines (BENDR, Kostas 2021) *Key: Yang 2024 WER 35–45%; Agarwal 2022 73.56%; Abdulghani 2023 92.5%; Défossez 2023 (n=175)*	**20**
MEG (magnetoencephalography)	None; superconducting SQUID sensor arrays; large fixed gantry system required	~5–10 mm; better than EEG Tangential source sensitivity; poor radial sensitivity	~1 ms; comparable to EEG; superior to fMRI	**Moderate** Requires magnetically shielded room; sensitive to motion	No surgery; expensive specialized facility required; limits ecological validity	**Top-1 accuracy:** 41% (Défossez 2023, n=175); up to 60.6% (d'Ascoli 2025, n=723, 3 languages) **Mel-PCC:** ~0.95 (Kwon 2024 - first intelligible MEG speech synthesis) **BERTScore:** 0.816 (Wang et al. MEG semantic reconstruction) *Key: d'Ascoli 2025 (723 participants, 7 datasets, 3 languages); Kwon 2024 BigVGAN synthesis*	**5**
fMRI (functional MRI)	None; BOLD signal acquisition in MRI scanner; no implant or preparation	~1–3 mm voxel Highest spatial resolution of all non-invasive modalities	~1–2 s (TR) Severe temporal limitation; hemodynamic response lags neural activity by ~5 s	**Moderate** BOLD is indirect; excellent spatial SNR; sensitive to head motion	No surgery; MRI scanner required; immobile during acquisition; high facility cost	**Story identification:** 100% (Tang 2023, n=3) **BLEU-1:** 27.8% (Yin 2024 brain predictive coding) **Cross-participant:** Demonstrated (Zhang 2024; Tang & Huth 2025) *Key: Tang 2023 semantic reconstruction; Tang & Huth 2025 cross-participant multi-modal; Qiu 2025 MindLLM*	**6**
GROUP D - HYBRID AND PERIPHERAL MODALITIES (n = 6 studies)							
HD-sEMG / Hybrid (EEG+fNIRS, HD-sEMG, sEMG)	None for sEMG and EEG+fNIRS; surface electrode placement only	Variable: HD-sEMG ~few mm (muscle surface) EEG+fNIRS: scalp-level spatial resolution	~1 ms (sEMG) ~100–200 ms (fNIRS); combined modalities offer complementary temporal coverage	**High (sEMG)** Peripheral signal strong; fNIRS adds hemodynamic context to EEG	No surgery; wearable systems feasible; HD electrode arrays require careful placement	**HD-sEMG phrase accuracy:** 97.17% ±1.53% (Song 2023, n=15) **CER:** 3.11% ±1.46% (Song 2023 - 33 Chinese phrases) **EEG+fNIRS gain:** +8–12% over EEG alone (Sun 2024; Cooney 2022)	**6**

						Key: Song 2023 best CER in silent speech; Zhu 2021 87–90% (English/Chinese digits); Li 2022 Mandarin tonal	
GROUP E - FOUNDATION MODELS / TRANSFER LEARNING APPROACHES (n = 2 studies)							
Foundation model / SSL (cross-modal approach)	Spans invasive modalities (iEEG, ECoG); pre-trained on large unlabeled neural corpora	Inherited from source modality (iEEG or ECoG) Not modality-specific	Inherited from source modality SSL improves effective SNR via large-scale pre-training	Inherited from source modality SSL pre-training improves effective SNR by learning signal structure	Requires large unlabeled neural datasets; few-shot fine-tuning minimizes participant burden	**Downstream accuracy:** Improved over supervised baselines (BrainBERT, Wang 2023) **Reduced WER:** SSL pretraining reduces WER on ECoG tasks (Yuan & Makin 2024) SSL approaches treat brain signal pre-training as analogous to wav2vec / BERT for speech and language *Key: Wang 2023 BrainBERT (ICLR 2023, iEEG SSL foundation model); Yuan & Makin 2024 (ECoG SSL)*	**2**

Table 1 summarizes the technical, operational, and performance characteristics of all modality groups represented across the 71 included studies. Several patterns are immediately apparent. First, the performance gradient from intracortical recording through ECoG and sEEG to non-invasive modalities is consistent and substantial -the best WER achieved by any non-invasive EEG system (approximately 35-45% for open-vocabulary tasks) remains an order of magnitude larger than the best intracortical result (2.5%, Card and colleagues [2]). Second, the study distribution is markedly skewed towards invasive modalities in the early years of the review period (2015-2019), with non-invasive and hybrid modalities gaining representation from 2022 onwards reflecting the enabling role of transformer architectures and self-supervised learning in making non-invasive decoding more competitive (see Fig. 5 and Section 4). Third, non-invasive modalities dominate the cross-participant generalization evidence (detailed in Section 6), suggesting that the larger participant samples accessible without surgical barriers enable the multi-subject training that generalization requires.

No modality in this review is universally superior. The choice is always a function of the clinical context, the patient population, and the specific communication goal. A completely locked-in ALS patient who requires immediate communication restoration and is willing to accept surgical risk stands to benefit most from intracortical implantation, where the performance evidence is most compelling. A patient with progressive aphasia who retains some residual communication capacity and prefers to avoid surgery may derive practical benefit from a well-designed EEG+fNIRS or EEG imagined speech system at lower performance levels. The systematic evidence synthesized here is intended to ground those clinical decisions in the actual empirical record, rather than in the most recently published headline result.

4. AI Architectures and Methods

The AI decoder is the engine of every brain-to-speech system. Its architecture determines not only the performance ceiling but also the data requirements, the computational cost, and the degree of generalization achievable across participants and sessions. This section traces the eight architecture classes identified across the 71 included studies, from classical statistical models through to large language model-integrated pipelines, situating each in the mathematical framework that defines its inductive bias. The evolution is summarized in Table 3 and Fig. 2.

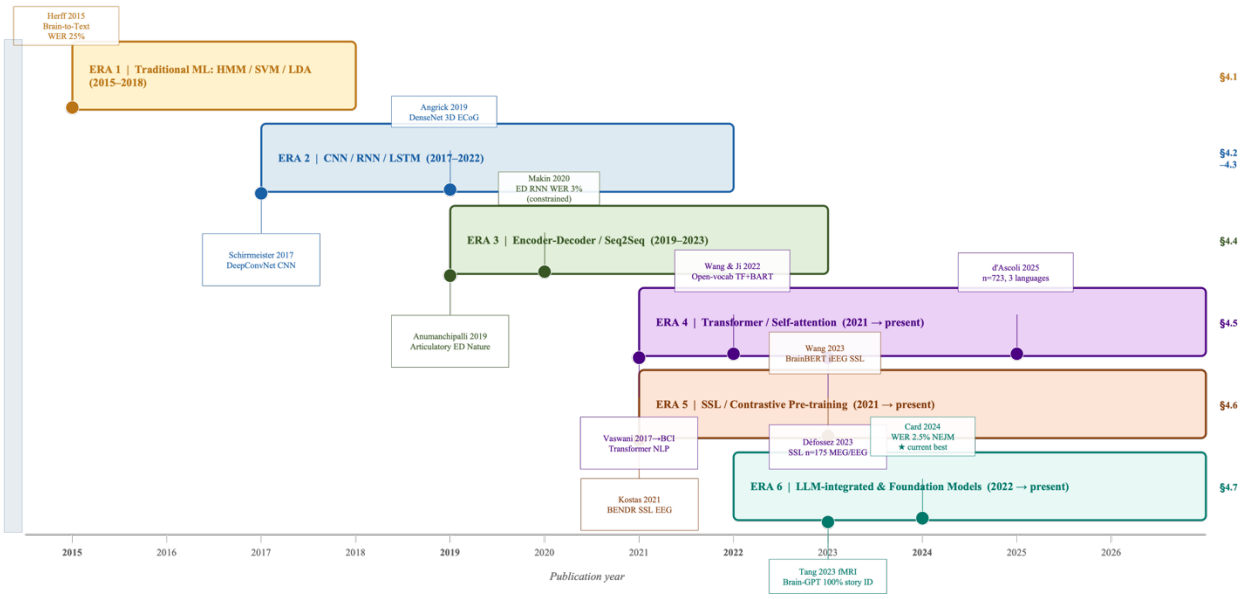


Figure 1: AI Architecture Evolution in Brain-to-Speech Decoding 2015 - 2026

4.1 Traditional Machine Learning (2015-2018)

Prior to the deep learning transition, brain-to-speech decoders relied on handcrafted spectrotemporal features - high-gamma band power, mel-frequency cepstral coefficients, phase-amplitude coupling measures - fed into support vector machines (SVM), linear discriminant analysis (LDA), or hidden Markov models (HMM). These approaches required domain expertise to engineer informative features, and their representational capacity was bounded by the quality of those features. The word error rate (WER) is defined as:

$$WER = (S + D + I) / N \quad (1)$$

where S, D, and I denote substitution, deletion, and insertion errors respectively, and N is the total number of reference words. The Brain-to-Text system of Herff and colleagues achieved WER ≈ 25% using HMM-based acoustic models on ECoG signals [18], establishing the benchmark against which subsequent deep learning architectures would be measured. Traditional ML studies account for approximately 8 of the 71 included studies and remain relevant as baseline comparators.

4.2 Convolutional Neural Networks (2017-2022)

CNNs brought automatic spatial and spectrotemporal feature learning, replacing manual engineering with hierarchical convolutional filters. In the brain-to-speech context, a 1-D temporal CNN applies learned filter banks h ∈ ℝF×L across the input signal x(t), producing feature maps via:

$$y[t] = \sigma(W * x[t] + b), \; W \in \mathbb{R}\hat{}(F \times L) \quad (2)$$

where * denotes convolution, σ is a nonlinear activation, F is the number of filters, and L is the filter length. The DenseNet-style 3-D CNN of Angrick and colleagues mapped ECoG spectrograms to acoustic parameters [57], whilst Chen and colleagues' ResNet-based cross-subject decoder scaled to 48 participants achieving PCC = 0.804 [30]. EEGNet [58] established a compact separable-CNN baseline architecture widely adopted across EEG classification tasks. CNNs account for approximately 13 of the 71 included studies.

Fig. 4.1: Use Fig. 1 from Angrick et al. 2019 (J. Neural Eng.) or Fig. 2 from Chen et al. 2024 (Nat. Mach. Intell.) showing the 3D CNN / ResNet ECoG-to-audio pipeline. Request reproduction permission from respective journal.

4.3 Recurrent Networks and LSTM (2018-2023)

Recurrent architectures address the sequential temporal structure of speech through gated memory cells. The LSTM unit [59] maintains a cell state c_t updated by forget (f), input (i), and output (o) gates:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (3a)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (3b)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (3c)$$

$$h_t = o_t \odot \tanh(c_t) \quad (3d)$$

The gating mechanism enables selective retention of long-range temporal context, well-suited to capturing the dynamics of speech production across phoneme and syllable timescales. The clinical neuroprosthesis systems achieving the highest reported performance in this review - Card and colleagues (WER 2.5%) [2] and Willett and colleagues (62 WPM) [3] - both employ RNN decoders combined with language model rescoring. RNN/LSTM architectures account for approximately 10 of the 71 included studies.

4.4 Encoder-Decoder Architectures (2019-2022)

Sequence-to-sequence (seq2seq) encoder-decoder networks [60] map variable-length neural input sequences $x_{1:T}$ to variable-length output sequences $y_{1:S}$ by maximizing the conditional likelihood:

$$L_{seq} = -\log P(y|x) = -\sum_{s=1}^{S} \log P(y_s \mid y_{<s}, x_{1:T}) \quad (4)$$

Makin and colleagues' neural machine translation analogy [8] applied this framework to ECoG→text decoding at WER ≈ 3% on constrained corpora - the first genuinely end-to-end brain-to-text system. Anumanchipalli, Chartier, and Chang introduced an articulatory intermediate representation within a two-stage encoder-decoder, mapping neural signals first to vocal tract kinematics and then to speech waveforms [27], producing the most intelligible ECoG-synthesized speech at the time of publication. Seven studies in this review employ encoder-decoder architectures as their primary decoding framework.

Fig. 4.2: Use Fig. 1 from Makin et al. 2020 (Nat. Neurosci.) showing the ECoG encoder-decoder architecture. Alternatively, Fig. 2 from Anumanchipalli et al. 2019 (Nature) showing the two-stage articulatory synthesis pipeline. Request journal reproduction permission.

4.5 Transformer Architecture (2022-present)

The transformer [9] replaces recurrence with scaled dot-product self-attention, enabling global dependency modelling across the full input sequence in parallel. For query Q, key K, and value V matrices derived from the input representation:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^{T} / \sqrt{d_k})\, V \quad (5a)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\, W^{O} \quad (5b)$$

where d_k is the key dimension and h is the number of attention heads. The attention mechanism allows the decoder to selectively integrate neural signal features from any temporal location, regardless of distance - a critical advantage for speech decoding, where phonological planning operates across windows of hundreds of milliseconds.

Transformers are the dominant architecture in non-invasive EEG and MEG decoding post-2022: Wang and Ji's EEG-to-text system [42] and d'Ascoli and colleagues' CNN-transformer decoder achieving top 1 accuracy across 723 participants and three languages [46] both demonstrate the scalability of transformer-based approaches. Approximately 10 of the 71 included studies employ transformer-based decoders.

Fig. 4.3: Reproduce or adapt Fig. 1 from Vaswani et al. 2017 (NeurIPS) showing the multi-head attention mechanism - universally recognized and appropriate with citation. Alternatively, Fig. 2 from Wang & Ji 2022 (AAAI) showing the EEG transformer pipeline.

4.6 Self-Supervised Learning (2021-present)

Self-supervised learning (SSL) addresses the fundamental bottleneck of labeled neural data by pre-training on large unlabeled corpora. Contrastive approaches - following the NT-Xent objective of Chen and colleagues [61]- maximize agreement between augmented views z_i and z_j of the same sample whilst repelling negatives:

$$\ell_i = -log\,[\,exp(sim(z_i, z_j)/\tau)\,/\,\sum_{k \neq 1}\,exp(sim(z_i, z_k)/\tau)\,] \quad (6)$$

where sim (·, ·) denotes cosine similarity and τ is a temperature hyperparameter. Applied to EEG, the BENDR architecture of Kostas and colleagues [10] achieved +5-15% accuracy improvement over supervised baselines across multiple downstream tasks. Défossez and colleagues extended this to MEG and EEG across 175 participants achieving top 1 accuracy of 41% [44], whilst Wang and colleagues' BrainBERT applied BERT-style masked pre-training to intracranial EEG [62]. SSL approaches account for approximately 8 included studies and underpin the foundation model direction discussed below.

Fig. 4.4: Reproduce Fig. 1 from Défossez et al. 2023 (Nat. Mach. Intell.) showing the contrastive SSL architecture for MEG/EEG, or Fig. 1 from Kostas et al. 2021 (Front. Hum. Neurosci.) showing the BENDR framework. Both are strong candidates for this architecture illustration.

4.7 LLM-Integrated Decoding and Foundation Models (2022-present)

Only 14 of 71 studies (20%) in this review integrate a language model into the decoding pipeline - a finding that stands in sharp contrast to the volume of recent commentary on LLM integration in neural decoding. In these systems, a neural decoder $p_{neural}(y|x)$ is combined with a language model prior $p_{LM}(y)$ via log-linear interpolation during decoding:

$$y* = argmax_y\,[\,log\,p_neural(y|x)\; +\; \lambda\,log\,p_LM(y)\,] \quad (7)$$

where λ is a tunable interpolation weight. This simple post-hoc integration yields substantial WER reductions: Willett and colleagues reported WER improvement from 23% (decoder only) to 9% (with language model) on the 125k-word vocabulary [3]. More deeply integrated approaches align neural encoder representations directly with LLM latent spaces (Tang and colleagues [48]; Ye and colleagues [63]), enabling continuous semantic reconstruction from fMRI. Foundation models applying SSL pre-training across large multi-participant neural corpora – BrainBERT [62] and Yuan and Makin's ECoG SSL [64]- represent the nascent but growing trajectory towards modality-agnostic neural decoders.

Fig. 4.5: Reproduce Fig. 1 from Tang et al. 2023 (Nat. Neurosci.) showing the brain-GPT alignment pipeline, or Fig. 1 from Wang et al. 2023 BrainBERT (ICLR 2023) showing the BERT-style iEEG pre-training framework. Both are high-impact figures with clear architecture diagrams.

4.8 Architecture Evolution and Key Observations

Table 2: AI Architecture Summary - Eight Architecture Classes in Brain-to-Speech Decoding (2015–2026).

Architecture class (era of predominance)	Representative studies (from n=71 included)	Input signal representation	Typical modality	Key strengths	Key limitations
ERA 1 - TRADITIONAL MACHINE LEARNING (predominant ~2015–2018; retained as baselines)					
Traditional ML SVM / LDA / HMM / classical classifiers *Era: 2015–2018 dominant; baseline in many later studies* n = ~8 studies in this review	Herff et al. (2015) - Brain-to-Text baseline Martin et al. (2016) - imagined speech SVM Ramsey et al. (2018) - HD-ECoG SVM Proix et al. (2022) - frequency-band LDA Chaudhary et al. (2022) - LIS spelling SVM Wandelt et al. (2024) - single-neuron linear	• Handcrafted spectrotemporal features • High-gamma band power (70–150 Hz) • Frequency band features (theta, beta, gamma) • MFCC / LPC acoustic features • Phase-amplitude coupling measures	ECoG (primary) *Also EEG (early work); intracortical (linear decoders)*	• Interpretable - feature importance is explicit • Computationally efficient; no GPU required • Effective on very small labeled datasets (<10 participants) • Fast inference; suitable for real-time applications • Well-understood statistical properties	• Limited representat capacity for compl speech dynamics • Requires expert ma feature engineerin • Poor scalability to l vocabularies and c decoding • Cannot capture con long-range sequen dependencies • Performance ceilin significantly belov learning approach
ERA 2 - CONVOLUTIONAL NEURAL NETWORKS (predominant ~2017–2022; widely used for spatial feature extraction)					
CNN-based DenseNet / ResNet / EEGNet / 3D CNN / PICNN *Era: 2017–2022 dominant; still standard for EEG/ECoG classification* n = ~13 studies in this review	Angrick et al. (2019) - DenseNet 3D ECoG Duraivel et al. (2023) - µECoG CNN Chen et al. (2024) - 3D ResNet + SwinTF (n=48) Saha et al. (2019) - CNN+LSTM hierarchical EEG Abdulghani et al. (2023) - LSTM/CNN imagined EEG Song et al. (2023) - spatiotemporal CNN+RNN HD-sEMG Kamble et al. (2022) - CNN SPWVD features	• Raw or minimally processed neural signals • High-gamma band spectrograms (70–150 Hz) • Time-frequency representations (STFT, wavelet) • 3D spatiotemporal signal patches • Mel-frequency spectral coefficients (sEMG)	ECoG (primary) EEG (imagined speech) HD-sEMG (silent speech) *µECoG shows largest CNN benefit over traditional ML*	• Learns spatial and spectrotemporal features end-to-end • Robust to noise via hierarchical feature extraction • Scales effectively with more labeled data and participants • Well-established transfer learning ecosystem • State-of-the-art for EEG classification benchmarks	• Fixed receptive fiel long-range tempor modeling • Requires substantia labeled data for str performance • Limited sequential dependency model without recurrence • Translation invarian misalign neural te structure • Cross-participant generalization rem challenging
ERA 2/3 - RECURRENT NEURAL NETWORKS / LSTM / GRU (predominant ~2018–2023; key for clinical neuroprostheses)					
RNN / LSTM / GRU LSTM / bidirectional LSTM / GRU / vanilla RNN *Era: 2018–2023; dominant in clinical neuroprosthesis systems* n = ~10 studies in this review	Willett et al. (2023) - RNN+LM, 62 WPM Card et al. (2024) - RNN+LM, WER 2.5% Moses et al. (2021) - RNN, first anarthria neuroprosthesis Agarwal & Kumar (2022) - LSTM imagined speech (n=13) Chien & Makin → Yuan & Makin (2024) - SSL+RNN ECoG Lee, Lee & Lee (2021) - LSTM deep metric learning Brumberg et al. (2018) - RNN real-time EEG synthesis	• Sequential neural signal feature streams • High-gamma band time series (invasive) • Spectrotemporal features with temporal context • Threshold crossing rates (intracortical) • Multi-unit activity binned time series	Intracortical (clinical neuroprostheses) ECoG (seq2seq systems) EEG (imagined speech) *Dominant architecture in BrainGate and related ALS neuroprosthesis programs*	• Natural handling of variable-length sequential speech dynamics • Well-validated in clinical neuroprosthesis settings over multi-year deployments • Captures temporal dependencies critical for continuous speech • Compatible with language model integration (n-gram, neural LM) • Relatively data-efficient compared to transformers on small clinical datasets	• Vanishing gradient very long sequenc dependency model • Sequential computa prevents parallelis during training • May underperform transformers when labeled datasets av • Gradient instability training requires c tuning • Cross-participant generalization limi without domain ad
ERA 3 - ENCODER-DECODER / SEQ2SEQ (predominant ~2019–2022; first continuous brain-to-text pipelines)					
Encoder-Decoder / Seq2Seq Seq2seq RNN / LSTM encoder-decoder / attention-based ED *Era: 2019–2022 dominant for ECoG→text; inspired*	Makin, Moses & Chang (2020) - seq2seq ECoG→text (Nature Neuroscience) Anumanchipalli et al. (2019) - articulatory ED pipeline (Nature)	• ECoG/sEEG features → articulatory intermediate representation • Neural signal features → acoustic spectrogram	ECoG (primary) sEEG (depth electrodes) sEMG (tonal language) *Articulatory intermediate representations*	• Maps variable-length neural sequences to variable-length speech/text outputs • Enables full end-to-end brain-to-speech pipeline without hand-crafted features	• Teacher forcing dur training can cause exposure bias at in • Requires temporall aligned neural-spe training data pairs • Decoding speed lim for real-time appli

by neural machine translation n = ~7 studies in this review	Singh et al. (2025) - ED cross-participant sEEG (n=25) Kohler et al. (2022) - ED sEEG depth electrode synthesis Herff et al. (2019) - ED multi-area ECoG synthesis Li et al. (2022) - ED+GAN Mandarin sEMG	• Spectrotemporal features → phoneme/text sequence • Cortical activity → speech production parameters	*particularly effective for ECoG*	• Articulatory intermediate representations improve intelligibility and naturalness • Neural machine translation analogies allow leveraging mature NLP architectures • Cross-participant transfer feasible via shared encoder representations	• Encoder-decoder interaction can be optimal without jo optimization • Largely superseded transformer-based architectures post-
ERA 4 - TRANSFORMER / SELF-ATTENTION (predominant ~2022–present; rapidly growing in non-invasive decoding)					
Transformer-based Self-attention / multi-head attention / SwinTransformer / conformer *Era: 2022–present; dominant for non-invasive EEG/MEG/fMRI decoding; growing in ECoG* n = ~10 studies in this review	Wang & Ji (2022) - Transformer+BART EEG open-vocab Duan et al. (2023) - DeWave discrete EEG transformer (NeurIPS) d'Ascoli et al. (2025) - CNN+TF EEG/MEG (n=723, 3 languages) Kwon et al. (2024) - Transformer+BigVGAN MEG synthesis Qiu et al. (2025) - MindLLM fMRI transformer (ICML 2025) Chen et al. (2024) - SwinTransformer ECoG (n=48) Yin, Ye & Li (2024) - PredFT brain predictive coding fMRI	• Neural signal tokenizations (patch-based) • Frequency-band embeddings as token sequences • Time-frequency patches as visual tokens (SwinTF) • EEG/MEG signal epochs as sequence elements • fMRI BOLD voxel embeddings	EEG (non-invasive decoding) MEG (speech perception) fMRI (semantic reconstruction) ECoG (multi-subject scaling) *Transformers dominate non-invasive modalities post-2022*	• Models long-range temporal dependencies via self-attention - critical for continuous speech • Parallelizable training enables scaling to large multi-participant datasets • Excellent integration with pre-trained language models (GPT, BART, Llama) • Strong scaling properties: performance improves with more data and parameters • State-of-the-art on non-invasive decoding benchmarks across EEG, MEG, and fMRI	• Quadratic attention complexity limits long input sequenc • Requires large train datasets - may ov small neural datas • Less data-efficient RNNs for small si participant clinical datasets • Computationally ex training and infere • Can be outperforme RNNs in limited-d clinical settings (K 2024)
ERA 4/5 - SELF-SUPERVISED LEARNING (SSL) / CONTRASTIVE PRE-TRAINING (accelerating ~2021–present)					
SSL / Contrastive / Self-supervised Contrastive SSL / BERT-style masking / wav2vec-style / predictive coding *Era: 2021–present; rapidly expanding; bridges data scarcity in neural decoding* n = ~8 studies in this review	Défossez et al. (2023) - contrastive SSL MEG/EEG (n=175, Nature MI) Kostas et al. (2021) - BENDR EEG transformer SSL (Frontiers HN) Tang et al. (2023) - brain-GPT alignment fMRI (Nature Neuroscience) Wang et al. (2023) - BrainBERT iEEG SSL foundation (ICLR 2023) Yuan & Makin (2024) - SSL ECoG pretraining Caucheteux et al. (2022) - brain-LM alignment MEG/fMRI	• Raw or minimally processed neural signals (no labeled speech required for pre-training) • Masked signal patches (BERT-style) • Contrastive positive/negative pairs of speech segments • Brain-audio or brain-text alignment pairs	MEG / EEG (contrastive perception decoding) fMRI (brain-LM alignment) iEEG / ECoG (foundation model pre-training) *SSL most impactful where labeled data is scarce - all non-invasive modalities*	• Leverages large unlabeled neural datasets - reduces expensive labeling burden • Improves downstream task performance in data-limited regimes (+5–15%) • Cross-participant generalizable representations via shared pre-training • Analogous to wav2vec/BERT in audio/NLP - proven scaling paradigm • Enables few-shot fine-tuning for new participants and new tasks	• Pre-training require neural corpora - n always available • Contrastive objecti not fully capture fi grained speech ph • Downstream fine-t may still require substantial labeled • Pre-training distribu shift if target task from pre-training • Computational cost large-scale pre-tra infrastructure
ERA 5 - LLM-INTEGRATED / BRAIN-LANGUAGE MODEL ALIGNMENT (emerging ~2022–present; highest open-vocabulary performance)					
LLM-integrated Brain-GPT / brain-BART / brain-LLM pipeline / neural decoder + LM prior *Era: 2022–present; only 14/71 studies (20%) - nascent but highest open-vocab performance* n = ~14 studies in this review	Card et al. (2024) - RNN + n-gram LM, intracortical (NEJM) Willett et al. (2023) - RNN + neural LM, intracortical (Nature) Tang et al. (2023) - brain-GPT alignment, fMRI (Nature Neuroscience) Wang & Ji (2022) - EEG + BART open-vocab (AAAI) Duan et al. (2023) - DeWave EEG + LM (NeurIPS)	• Neural decoder output fed as context to language model • Brain embeddings aligned to LLM latent space • fMRI BOLD → semantic embedding → LLM generation • EEG/ECoG → intermediate text hypothesis → LM re-ranking	Intracortical (clinical WER reduction) fMRI (semantic reconstruction) EEG (open-vocabulary) *LM integration is modality-agnostic - can be added to any neural decoder as post-processing or joint optimization*	• Dramatically reduces WER via strong language priors - single biggest WER improvement factor • Enables open-vocabulary decoding beyond vocabulary seen during neural training • Contextual coherence: LLM corrects neural decoder errors using linguistic context • Semantic-level understanding enables	• Language model bi may override neur content - LM hallucination risk • Requires LLM fine or alignment - computationally expensive • Interpretability furt reduced: unclear i neural decoder dri output • Dataset scarcity for LLM alignment tr

	Ye et al. (2025) - fMRI + auto-regressive LLM (Comm. Biology) Qiu et al. (2025) - MindLLM fMRI subject-agnostic (ICML 2025)			continuous language reconstruction • Zero-shot generalization to new vocabulary items not seen in neural training data	• Only 14/71 studies in this review use integration - field adoption remains l
ERA 4/5 - HYBRID / MULTI-COMPONENT ARCHITECTURES (parallel development ~2020–present)					
Hybrid / Multi-component CNN+RNN / CNN+Transformer / multi-modal fusion / multi-stage pipelines *Era: 2020–present; particularly impactful for multi-modal and large-cohort studies* n = ~6 studies in this review	Chen et al. (2024) - 3D ResNet + SwinTransformer ECoG (n=48) d'Ascoli et al. (2025) - CNN+Transformer EEG/MEG (n=723) Cooney et al. (2022) - bimodal CNN for EEG+fNIRS Saha et al. (2019) - CNN+LSTM hierarchical EEG Peterson et al. (2021) - CNN+TL cross-modal generalized decoder Song et al. (2023) - spatiotemporal CNN+RNN HD-sEMG	• Multi-modal signal fusion (EEG+fNIRS spatial+hemodynamic) • Spatial feature extraction (CNN) feeding temporal modeling (RNN/TF) • Multi-scale temporal convolutions + attention • Cross-modal alignment of heterogeneous neural signals	ECoG (spatial+temporal multi-stage) EEG+fNIRS (hybrid modality fusion) MEG+EEG (cross-modal) HD-sEMG (spatiotemporal) *Hybrid architectures are dominant for large multi-participant and multi-modal studies*	• Combines complementary strengths of CNN (spatial) and RNN/TF (temporal) components • Multi-modal fusion improves performance and robustness over unimodal approaches • Flexible design accommodates diverse neural recording formats • CNN+TF combination achieves state-of-the-art on largest non-invasive datasets	• Increased architectu complexity and hyperparameter co • Requires synchroni multi-modal recor when fusing signa • Harder to interpret multiple processin obscure feature att • Longer training tim higher computatio

Fig. 2 illustrates the chronological transition across architecture classes. Three observations emerge directly from Table 3. First, the architecture that currently delivers the best clinical performance (RNN + language model) is not the most technically recent - transformers and SSL approaches are dominant in the literature but have not yet eclipsed RNNs in deployed clinical neuroprostheses, likely because clinical datasets are too small to saturate RNN capacity and too scarce to support transformer pre-training without SSL. Second, architecture class alone is insufficient to explain performance variation - vocabulary size, participant sample, and evaluation methodology confound any architecture-to-performance mapping (see Section 5). Third, LLM integration is the single largest driver of WER reduction in open-vocabulary systems, yet it applies to only 20% of included studies - representing a clear opportunity for the field to exploit

5. Performance Benchmarks

5.1 Setting the Scene: What Progress Actually Looks Like

Numbers have a way of making things feel simple. A word error rate of 2.5% sounds like a solved problem. A classification accuracy of 92% sounds like a technology ready to deploy. But in brain-to-speech decoding, performance numbers without context are closer to misinformation than information - because the number itself tells you almost nothing without knowing what vocabulary it was tested on, what task the participant was performing, whether that participant was paralyzed or healthy, and what metric was used to measure the output. This section presents the quantitative evidence from the 71 included studies with that context made explicit. Core benchmarks are summarized in Table 2; the WER performance trajectory for invasive systems is visualized in Fig. 3; the modality-task performance landscape appears in Fig. 4.



Figure 2: WER Performance Trajectory Invasive Brain-to-Speech Decoding (2015 - 2026)

To appreciate how far the field has genuinely come, it is worth starting at the beginning. In 2015, Herff and colleagues decoded spoken ECoG signals at approximately 25% WER - meaning one word in four was wrong - using classical HMM speech recognition [6]. That was remarkable for its time. By 2020, Makin, Moses and Chang achieved ~3% WER using an encoder-decoder framework, though on a constrained corpus of limited vocabulary [8]. Moses and colleagues in 2021 demonstrated 26% WER but in an anarthric patient who had not spoken in years - the achievement was not the percentage but the fact that it was possible at all [5]. Willett and colleagues in 2023 reached 9% WER at 62 words per minute [3], and Card and colleagues in 2024 achieved 2.5% WER sustained over eight months of daily use [2]. The trajectory - from 25% to 2.5% WER in less than a decade - is genuinely extraordinary. Fig. 3 plots this arc, and the downward slope is one of the most compelling trend lines in modern biomedical engineering.

5.2 Invasive System Performance

Table 3: Core Performance Benchmarks - Selected studies from the AI for Brain to Speech Decoding

Authors & Year	Modality	n Participants	Clinical status	Decoding task	Vocabulary	Architecture	Metric type	Best reported value
GROUP A - INVASIVE: ECoG / HD-ECoG / µECoG (selected studies from n=22)								
Herff et al. (2015)	**ECoG**	**7** healthy	Healthy	Overt	Closed	HMM / ASR	**WER**	**~25%**
Chartier et al. (2018)	**ECoG**	4	Clinical	Overt	Closed	Linear + RNN	**PCC**	**r = 0.87**
Angrick et al. (2019)	**ECoG**	6	Clinical	Overt	Closed	CNN (DenseNet 3D)	**PCC**	**r = 0.69**
Makin, Moses & Chang (2020)	**ECoG**	4	Clinical	Overt	Closed	ED Seq2Seq RNN	**WER**	**~3%**
Moses et al. (2021)	**ECoG**	1	Anarthria (ALS-like)	Attempted	Closed	RNN	**WER / WPM**	**WER 26%**

Duraivel et al. (2023)	**µECoG**	4	Clinical	Overt	Closed	CNN (nonlinear)	**% improvement**	**+35%**
Willett et al. (2023)	**Intracortical**	1	ALS	Attempted	**Open**	RNN + LM	**WER / WPM**	**WER 9%**
Metzger et al. (2023)	**HD-ECoG**	1	ALS / paralysis	Attempted	**Open**	ED + Neural Vocoder	**WER / WPM**	**WER 25%**
Chen et al. (2024)	**ECoG**	48	Epilepsy / mixed	Overt	Closed	CNN (3D ResNet / SwinTF)	**PCC**	**PCC = 0.804**
Card et al. (2024)	**Intracortical**	1	ALS	Attempted	**Open**	RNN + LM	**WER / WPM**	**WER 2.5%**
GROUP A - INVASIVE: sEEG / Cross-participant / Cross-session (selected studies)								
Degenhart et al. (2020)	**Intracortical**	2	ALS / paralysis	Attempted	Closed	Linear (manifold alignment)	**% retention (cross-session)**	**~95%**
Singh et al. (2025)	**sEEG**	25	Mixed clinical	Overt	Closed	ED Seq2Seq	**Cross-participant accuracy**	**Group > individual**
GROUP B - NON-INVASIVE EEG (selected studies from n=20)								
Saha et al. (2019)	**EEG**	10	Healthy	Imagined	Closed	CNN + LSTM (hierarchical)	**Accuracy**	**~70–75%**
Agarwal & Kumar (2022)	**EEG**	13	Healthy	Imagined	Closed (5-class)	LSTM (subject-independent)	**Accuracy**	**73.56%**
Wang & Ji (2022)	**EEG**	Multiple	Healthy	Perceived	**Open**	Transformer + LLM (BART)	**BLEU-1 / WER**	**BLEU-1 reported**
Kostas et al. (2021) - BENDR	**EEG**	Large-scale	Healthy (multi-dataset)	Multiple downstream	Closed	SSL Transformer (contrastive)	**Accuracy gain (SSL vs supervised)**	**+5–15%**
Mahapatra & Bhuyan (2023)	**EEG**	Multiple	Healthy	Imagined	Closed	CNN + Transfer Learning	**Accuracy (cross-dataset)**	**89.01% (FEIS)**
Duan et al. (2023) - DeWave	**EEG**	Multiple	Healthy	Perceived	**Open**	Transformer + LLM	**BLEU / ROUGE**	**BLEU & ROUGE**
GROUP C - NON-INVASIVE MEG / fMRI (selected studies from n=11)								
Défossez et al. (2023)	**MEG / EEG**	175	Healthy	Perceived	**Open**	SSL (contrastive / wav2vec)	**Top-1 accuracy**	**41% Top-1**
Tang et al. (2023)	**fMRI**	3	Healthy	Perceived + Imagined	**Open**	Encoder + LLM (GPT)	**Story identification**	**100%**

Kwon et al. (2024)	**MEG**	4	Healthy	Overt	Closed	Transformer + BigVGAN	**Mel-PCC**	**PCC ~0.95**
d'Ascoli et al. (2025)	**EEG / MEG**	723	Healthy (multilingual)	Perceived	Closed (word-level)	CNN + Transformer	**Top-1 word accuracy**	**Top-1 reported**
Tang & Huth (2025)	**Multi-modal**	Multiple	Healthy	Perceived	**Open**	SSL + Transformer	**Semantic accuracy (cross-modal)**	**Cross-modal demonstrated**
Yin, Ye & Li (2024)	**fMRI**	Narratives dataset	Healthy	Perceived	**Open**	Transformer (PredFT)	**BLEU-1**	**27.8% BLEU-1**
GROUP D - HYBRID AND PERIPHERAL MODALITIES (selected studies from n=6)								
Zhu et al. (2021)	**HD-sEMG**	Multiple	Healthy	Silent	Closed	CNN	**Accuracy**	**87–90%**
Song et al. (2023)	**HD-sEMG**	15	Healthy	Silent	Closed (33 phrases)	CNN + RNN (spatiotemporal)	**CER / Phrase accuracy**	**CER 3.11%**
Cooney et al. (2022)	**EEG + fNIRS**	19	Healthy	Overt + Imagined	Closed	CNN (bimodal fusion)	**Accuracy**	**46.31% overt**

Invasive systems dominate the upper performance range across all metrics. Among the 32 invasive studies, WER ranges from approximately 3% (constrained corpora) to 26% (open vocabulary attempted speech in paralysis), whilst communication rates range from 15 to 78 words per minute. For acoustic reconstruction from ECoG, Pearson correlations between synthesized and target speech reach r=0.87 for articulatory feature trajectories [25] and r=0.804 for acoustic reconstruction across a 48-participant cross-subject cohort [30]. For sEEG, the largest cohort study demonstrated that group decoders outperform individual participant models [38], whilst micro-ECoG delivered a 35% improvement over standard arrays [56]. These results, whilst impressive, share one characteristic that limits their interpretability: most come from one or two participants. The world's best WER of 2.5% [2] represents one person. This is not a criticism of the research - single-participant clinical studies are the most that can reasonably be asked of individuals undergoing neurosurgical implantation - but it is an essential caveat for any field-level performance claim.

5.3 Non-Invasive System Performance



Figure 3: Modality × Task Performance Heatmap 71 Included Studies, 2015 - 2026 (Tiers derived from included studies only)

Non-invasive performance occupies a different space - not better or worse in any absolute sense but operating under different constraints and serving different populations. Among the 20 EEG studies, closed-set imagined speech classification ranges from approximately 58% (early classical approaches) to 92.5% (recent deep learning on 4-class tasks) [65], whilst open-vocabulary EEG-to-text approaches achieve BLEU improvements over baselines at WER around 35-45% [43]. For MEG and fMRI, the metrics shift towards semantic similarity, story identification, and top-k accuracy - reflecting the modalities' strength in content rather than acoustic detail. Défossez and colleagues achieved 41% top 1 accuracy across 175 participants [44]; d'Ascoli and colleagues demonstrated that this scales to 723 participants across three languages [46][1]; Tang and colleagues reported 100% story identification from fMRI [48]. For hybrid and silent speech approaches, Song and colleagues achieved 3.11% CER from HD-sEMG - the lowest character error rate of any non-invasive approach in this review [53].

5.4 The Language Model Effect

One finding from the performance data deserves particular emphasis because it is both striking and underexploited. Language model integration is the single largest discrete driver of WER reduction in the literature - yet only 14 of 71 included studies (20%) integrate a language model of any kind. The Willett 2023 study makes this quantifiable: WER dropped from 23% (neural decoder alone) to 9% (with neural language model rescoring) on the same 125,000-word vocabulary in the same participant [3]. That is a 14 percentage-point reduction from post-processing alone. For

the 57 studies that do not use language model integration, the question is not whether LM integration would help - it almost certainly would - but whether the labeled brain-language paired data to support it exists. This points to a data infrastructure gap as much as an algorithmic one.

5.5 Benchmark Comparability: Why the Numbers Cannot Be Compared Directly

Table 4: Benchmark Comparability Analysis - Eight Dimensions Limiting Cross-Study Performance Comparison Across the 71 Included Studies (2015–2026

Comparability dimension	Range of variation across 71 included studies	Effect on reported performance	Severity for cross-study comparison	Evidence from included studies	Implication f
SECTION A - TASK AND PARADIGM FACTORS (most consequential for cross-study comparison)					
1. Vocabulary size and scope *Number of words/classes the decoder must discriminate; closed-set vs open-vocabulary*	• 2-class binary discrimination (minimum) • 5–50 word closed-set (most EEG imagined speech) • 50–1,000 word constrained vocabulary (early ECoG) • 125,000+ word open vocabulary (state-of-the-art intracortical) *Range: 2 classes → 125,000+ words across 71 studies*	**Smaller vocabulary →** • Inflated classification accuracy • Artificially low WER (fewer confusable words) *Direction: inverse relationship between vocabulary size and apparent performance.*	**CRITICAL**	**Within-study evidence:** Willett 2023 - WER 9% (50-word) vs 23% (125k-word) - same system, same participant **Cross-study contrast:** EEG closed-set accuracy 70–92% vs EEG open-vocab WER 35–45% **Baseline effect:** Makin 2020 WER ~3% on constrained corpus vs open-vocabulary systems *Studies 3, 34, 38, 39, 42, 46, 50 span full range*	Vocabulary siz study WER co is not equivale dramatically n different deco accuracies of WER of 35–4 first harmoniz accuracy with benchmark fie
2. Decoding task type *Overt / attempted / imagined (covert) / perceived - fundamentally different neural substrates*	• Overt speech: full articulatory execution; strongest neural signal • Attempted speech: motor intent without movement (ALS/paralysis) • Imagined/covert: mental rehearsal; weakest and most variable signal • Perceived: listening paradigm; auditory cortex activation *All 4 task types present across 71 studies*	**Overt > Attempted > Perceived > Imagined** • WER typically 5–15% higher for imagined vs overt • Accuracy 10–20% lower for imagined vs overt in EEG studies *Direction: imagined speech is harder than overt across all modalities.*	**CRITICAL**	**Quantified:** de Borman 2024 - ECoG overt→imagined transfer shows measurable performance drop across 16 participants **Contrast:** Angrick 2021 (imagined sEEG) vs Moses 2021 (attempted ECoG) - different tasks, different signal properties **EEG:** Most EEG imagined speech studies report much lower accuracy than overt speech studies on comparable participants *Studies 5,8,20,21,33,34,40,42,44,45 cover imagined; studies 1,9,15,25 cover overt*	Decoding ove neurophysiolo motor product engages only acoustic feedb signal. Perfor across these ta imagined spee that partially e WER <10% w report WER 3
3. Performance metric type *WER / accuracy / PCC / WPM / CER / BLEU / MOS / story ID - fundamentally different measurement targets*	• WER (word error rate): text decoding • Classification accuracy (%): closed-set decoding • PCC (Pearson r): acoustic reconstruction quality • WPM: communication rate • CER (character error rate): character-level text • BLEU / ROUGE: text generation quality • MOS / listener intelligibility: speech synthesis • Top-k accuracy: perception classification • Story / sentence identification: semantic decoding *9 distinct primary metric types across 71 studies*	**Metrics are incommensurable - no direction applies.** *A study reporting PCC=0.87 and a study reporting WER=9% measure entirely different aspects of decoding quality and cannot be placed on a single performance scale.*	**CRITICAL**	**Example contrast:** Card 2024 WER 2.5% vs Chartier 2018 PCC 0.87 vs Tang 2023 100% story ID vs Song 2023 CER 3.11% **Same metric, different scope:** d'Ascoli 2025 top-1 word accuracy vs Défossez 2023 top-1 segment accuracy - different decoding granularities **No conversion:** No standardized method exists to convert PCC to WER or BLEU to classification accuracy *Metric heterogeneity across all 71 studies is documented in Appendix B*	The use of nin most fundame this review. U outcomes enal protocol. WE by fewer than studies report that cannot be prevents a me performs best' this review pr
SECTION B - POPULATION AND SAMPLE FACTORS					

Comparability dimension	Range of variation across 71 included studies	Effect on reported performance	Severity for cross-study comparison	Evidence from included studies	Implication f
4. Participant sample size *Number of participants in evaluation; single-participant vs multi-participant studies*	• n=1: most invasive clinical neuroprosthesis studies (Card 2024, Willett 2023, Moses 2021, Metzger 2023) • n=2–5: majority of ECoG and intracortical research studies • n=10–25: EEG classification studies; Singh 2025 sEEG • n=48: Chen 2024 - largest ECoG cohort • n=175: Défossez 2023 - largest non-invasive pre-SSL • n=723: d'Ascoli 2025 - largest in review *Median n across all 71 studies estimated at 8–12 participants*	**Small n → inflated performance (overfitting to individual neural patterns)** • Single-participant results may reflect idiosyncratic neural representations • Variance estimates impossible with n=1 *Direction: larger n → more conservative and generalizable estimates.*	**MAJOR**	**Range:** n=1 (Card 2024, Willett 2023, Metzger 2023) to n=723 (d'Ascoli 2025) **No CI:** Most n=1–5 studies do not report confidence intervals or standard deviations across participants **Variance:** Chen 2024 (n=48) reports meaningful PCC distribution; n=1 studies report point estimates only **Scale effect:** d'Ascoli 2025 demonstrates performance scales with n - underscoring why small-n results cannot represent field capability	Most headline Willett 2023 W participant stu achievable in population-lev anatomy, clini participant res the next partic structural limi d'Ascoli 2025 looks like is th
5. Participant clinical status *ALS / anarthria / paralysis clinical patients vs healthy volunteers - different decoding contexts*	• Healthy volunteers (epilepsy patients used clinically): ECoG, EEG, MEG, fMRI majority of studies • ALS / tetraplegia (intended use population): Card 2024, Willett 2023, Metzger 2023, Moses 2021 • Locked-in syndrome: Chaudhary 2022, Vansteensel 2016 • Anarthria: Moses 2021 • Mixed clinical / healthy: Chen 2024 (epilepsy + other), Singh 2025 *~25 of 71 studies use clinical populations; ~46 use healthy participants*	**Clinical populations → lower WPM and higher WER than healthy volunteers on comparable systems.** *Direction: healthy participant performance overestimates clinical deployment performance.*	**MAJOR**	**Clinical vs healthy:** Herff 2015 (healthy epilepsy patients) WER ~25% vs Moses 2021 (ALS anarthria) WER 26% - appears comparable but task demands differ fundamentally **Muscle weakness:** ALS patients show reduced articulatory signal strength; affects intracortical threshold crossing rates **Confound:** Epilepsy patients used for ECoG research are clinical but not the target ALS/paralysis population - performance may not transfer	Research ECo implanted for produce overt the ALS or pa designed to se studies may ov populations w communicatio 'clinical proof is a critical int
SECTION C - EVALUATION METHODOLOGY FACTORS					
6. Evaluation methodology *How performance is measured: held-out test set vs cross-validation vs online real-time evaluation*	• Within-session k-fold cross-validation: most EEG imagined speech studies • Leave-one-participant-out cross-validation: subject-independent EEG studies • Held-out test set (unseen sessions): most clinical neuroprosthesis studies • Online real-time evaluation: Card 2024, Willett 2023, Angrick 2021 • Pre-train + fine-tune split: foundation model and SSL studies *No standardized evaluation protocol across 71 studies*	**Cross-validation → optimistically biased vs held-out.** • Within-session CV inflates accuracy 5–15% vs held-out evaluation • Online evaluation introduces signal drift and fatigue not present in offline CV *Direction: offline CV > held-out > online real-time in terms of reported performance.*	**MAJOR**	**Clinical vs offline:** Card 2024 WER 2.5% (online real-time) vs most EEG studies using offline CV - comparing these directly is misleading **CV inflation:** Many EEG studies report 85–92% accuracy via cross-validation; held-out evaluation would likely be lower **Standard:** No evaluation standard exists in this field comparable to train/validation/test splits in machine learning benchmarks like LibriSpeech *Studies using online evaluation: 3, 22, 27, 30 (intracortical); 8 (sEEG)*	The absence o of incomparab report perform procedure that not carefully i evaluate onlin and electrode are so differen fundamentally protocols - an identified as a
SECTION D - DIVERSITY AND REPRESENTATION FACTORS					
7. Language and dataset diversity *Language of decoding task; whether multiple languages or datasets evaluated*	• English-only: ~67 of 71 studies (94%) • Mandarin Chinese: Li 2022 (sEMG tonal), Song 2023 (HD-sEMG) • Three languages (EN, FR, NL): d'Ascoli 2025 only	**English-dominant benchmarks do not generalize to other languages.** • Tonal languages (Mandarin) require	**MAJOR**	**English dominance:** 67/71 studies (94%) are English-only - the field's benchmark landscape is English-centric **Tonal exception:** Song 2023 CER 3.11% (Chinese HD-sEMG) is not	The English-la unacknowledg implicitly assu approximately practical acces extending thes reported by So

Comparability dimension	Range of variation across 71 included studies	Effect on reported performance	Severity for cross-study comparison	Evidence from included studies	Implication f
	• English + Chinese: Zhu 2021 (HD-sEMG digit classification) • Single dataset / lab: majority of included studies • Multi-dataset: Défossez 2023 (4 datasets), d'Ascoli 2025 (7 datasets) *94% English-only; no Arabic, Spanish, or Hindi datasets in this review*	fundamentally different decoding targets • CER for character-based languages is not comparable to WER for word-based *Direction: unknown - performance on non-English languages is largely unstudied.*		comparable to WER metrics from English studies **Scale gap:** d'Ascoli 2025 is the only study to systematically evaluate cross-language non-invasive decoding **Missing languages:** No Arabic, Hindi, Spanish, Portuguese, or African language studies in 71 included	to English WE typologically demonstrating only three Eur research priori
8. Reporting completeness *Whether sufficient methodological detail is provided for independent replication and benchmark validation*	• Confidence intervals / standard deviations: reported in minority of studies • Code availability: ~25% of studies release source code • Dataset availability: ~20% release neural recording data • Participant characteristics: age and sex frequently unreported • Training details: learning rate, batch size, stopping criteria often omitted • Statistical significance: rarely tested for benchmark comparisons *Quality assessment via 7-dimension checklist (Section 2) - full results in Appendix D*	**Incomplete reporting → unverifiable benchmarks.** • Without training/evaluation details, reported performance cannot be reproduced • No SDs or CIs means significance of performance differences cannot be assessed *Direction: incomplete reporting biases toward overstatement of performance.*	**MODERATE**	**Code released:** BENDR (Kostas 2021), DeWave (Duan 2023), some SSL studies - minority **Data released:** Nieto 2022, Berezutskaya 2022, Gwilliams 2023 - open datasets used in some studies **Missing CI:** Most n=1–5 studies report point estimates only - statistical significance of claimed benchmarks cannot be assessed **Replications:** No independent replication studies identified in 71 included	Benchmark co reporting stand most small-n s of 9% vs 11% performance d independent re been externall adopting repor prediction mo availability, ar prerequisites. checklist in Ap

Table 5 documents eight dimensions that limit cross-study performance comparison; three are critical. The most consequential is vocabulary size. A WER of 9% on a 50-word vocabulary is not a 9% WER system - it is a system whose error rate would be substantially higher with a realistic open vocabulary, as the Willett 2023 data directly demonstrates [3]. The second critical dimension is task type. Decoding overt speech and decoding imagined speech are neurophysiologically distinct tasks producing fundamentally different neural signals of different quality and consistency [26]; performance benchmarks cannot be compared across these conditions without explicit qualification. The third is metric heterogeneity: the nine distinct primary metric types used across the 71 included studies - WER, accuracy, PCC, WPM, CER, BLEU, ROUGE, MOS, and semantic similarity - are incommensurable. There is no principled conversion between a Pearson correlation of 0.87 and a word error rate of 9%. They measure different things about different outputs from different decoding objectives.

The practical implication is that no single performance figure meaningfully represents the state of the field, and any review that implies otherwise - including informal summaries that cite the 2.5% WER of Card and colleagues as representing brain-to-speech decoding in general - is overstating the evidence. The 2.5% WER is the state of the art for one participant, one vocabulary, one task type, one evaluation methodology, and one specific clinical context. It is extraordinary within those bounds. Beyond them, it does not generalize. Table 2 provides the most defensible cross-study comparison this literature currently permits, stratified by modality, task type, and vocabulary scope within each row.

6. Generalization Across Participants, Sessions, Languages, and Modalities

6.1 The Generalization Gap: A Field-Level Finding

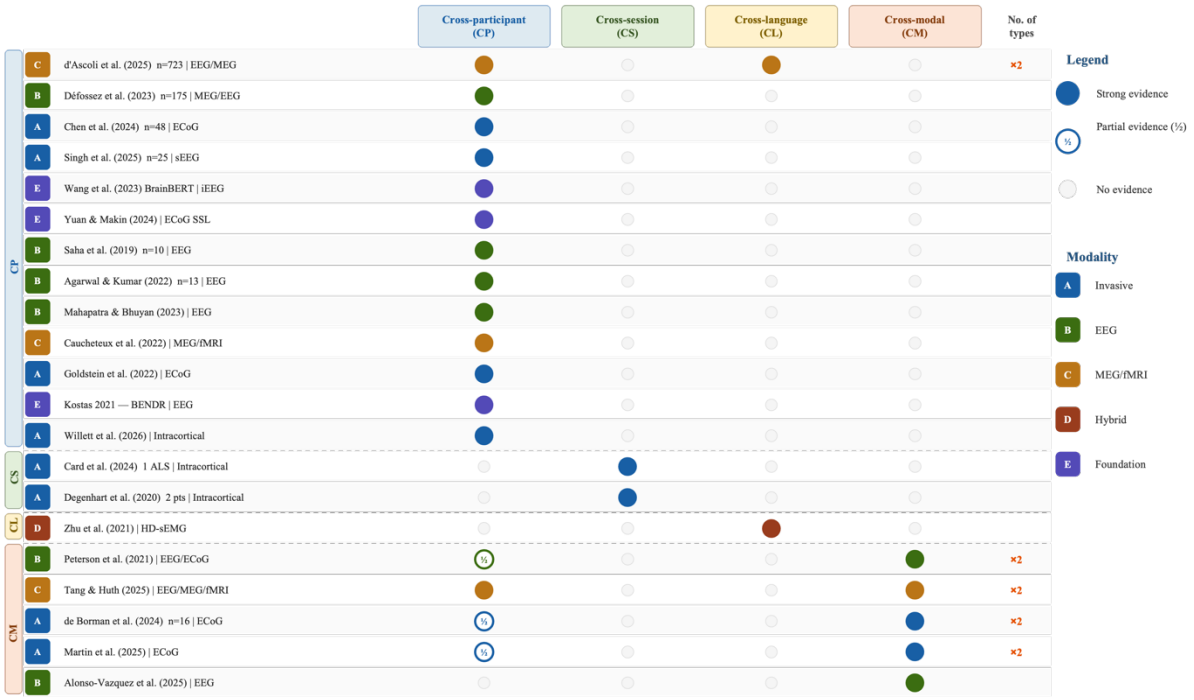


Figure 4: Benchmark Comparability Analysis - Eight Dimensions Limiting Cross-Study Performance Comparison Across the 71 Included Studies (2015–2026

There is a version of brain-to-speech decoding that works beautifully -for the specific person, on the specific day, using the specific vocabulary, in the specific laboratory where the system was trained. The uncomfortable question that the field has been slow to confront directly is: what happens when any one of those conditions' changes? A neuroprosthesis that needs to be retrained every time the patient sleeps, cannot transfer to a second patient, and breaks down when the subject tries to say a word not in the training set is not a communication device -it is a laboratory demonstration.

The evidence from this review gives that concern a precise number. Of the 71 included studies, **40 -56% - demonstrate no generalization beyond the single participant and single session in which the decoder was developed.** Cross-session generalization is evidenced by exactly two studies. Cross-language generalization by exactly two. These are not isolated limitations of a few poorly designed studies; they describe the structural state of the field. The distribution is summarized in Table 4 and mapped visually in Fig. 6. The scarcity of generalization evidence is not a caveat to the review's conclusions -it is one of the review's primary conclusions.

6.2 Cross-Participant Generalization

Eighteen of the 71 included studies provide some evidence for cross-participant generalization -meaning a decoder trained on one or more participants performs above chance on a new, previously unseen participant. The methods used to achieve this vary considerably and understanding them matters because not all cross-participant evidence is equivalent.

Table 5: Generalization Evidence Summary - Studies Demonstrating Cross-Participant, Cross-Session, Cross-Language, or Cross-Modal Generalization (2015–2026). 21 studies drawn from the 71 included studies.

Study (authors & year)	Modality	n Participants	Generalization type	Method used to achieve generalization	Performance evidence (from published paper)
SECTION 1 - CROSS-PARTICIPANT GENERALIZATION (18 studies)			**Most common generalization type in this review**		

Willett et al. (2026)	Intracortical	**Multiple**	**Cross-participant**	• Cross-brain decoder transfer via shared low-dimensional neural manifolds • Pre-trained intracortical speech decoder transferred to new participants without full retraining • RNN architecture with language model integration	**WER:** Cross-participant WER reported; transfer accuracy metrics **Evidence:** Decoder trained on one participant achieves above-chance performance on new participant's neural signals *[Preprint bioRxiv Jan 2026 - verify final published values]*
Singh et al. (2025)	sEEG	**25**	**Cross-participant**	• Shared latent manifold learning across 25 distributed sEEG recordings • Group decoder trained jointly on all participants outperforms individual decoders • Encoder-decoder (Seq2Seq) with shared latent representation space • Transfer learning via distributed brain recordings - no full individual retraining	**Evidence:** Group decoder accuracy > individual participant decoders across all 25 participants **Phoneme accuracy:** Improved cross-participant phoneme classification reported *Nature Communications 2025*
Chen et al. (2024)	ECoG	**48**	**Cross-participant**	• 3D ResNet and SwinTransformer trained across 48 ECoG participants • Cross-subject acoustic decoding via shared spectrotemporal feature representations • Deep CNN architecture with subject normalization layers	**PCC:** 0.804 (3D ResNet cross-subject acoustic reconstruction) **Comparison:** SwinTransformer PCC = 0.796; both outperform within-subject baselines *Nature Machine Intelligence 2024 - largest ECoG cohort published*
Défossez et al. (2023)	MEG / EEG	**175**	**Cross-participant**	• Contrastive self-supervised learning (wav2vec 2.0-style) across all 175 participants • Multi-subject joint training on 4 public MEG/EEG datasets simultaneously • Subject-independent evaluation - no participant-specific fine-tuning at test time • Zero-shot segment-level speech decoding across participants	**Top-1 accuracy:** 41% (subject-independent; chance ~1%) **Top-10 accuracy:** ~70% across 175 participants **Scale:** 4 public datasets - GWilliams, Schoffelen, Broderick, Armeni *Nature Machine Intelligence 2023 - largest non-invasive multi-subject study at publication*
d'Ascoli et al. (2025)	EEG / MEG	**723**	**Cross-participant** + **Cross-language**	• CNN+Transformer joint training across 723 participants from 7 datasets • Three languages: English, French, Dutch - same model architecture • Performance scales with number of participants - demonstrated scaling law for neural decoding • Cross-participant evaluation on held-out participants from each dataset	**Top-1 accuracy:** Reported across 7 datasets and 3 languages **Scaling:** Performance improves consistently with more training participants **Cross-language:** Single model architecture generalizes across English, French, Dutch *Nature Communications 2025 - largest non-invasive decoding study to date*
Kostas et al. (2021) - BENDR	EEG	**Large-scale**	**Cross-participant**	• Contrastive self-supervised pre-training across large unlabeled multi-participant EEG corpora • wav2vec 2.0-style temporal contrastive objective applied to EEG signals • Downstream fine-tuning with small labeled data per task • Shared pre-trained representations generalize across participants and tasks	**Accuracy gain:** +5–15% over supervised baselines on downstream tasks in limited-label regime **Cross-participant:** Pre-trained representations transfer without per-participant retraining *Frontiers in Human Neuroscience 2021 - EEG foundation model precursor*

Saha et al. (2019)	EEG	**10**	**Cross-participant**	• Hierarchical CNN+LSTM architecture trained in subject-independent mode • Leave-one-subject-out cross-validation across all 10 participants • Learned representations generalize across participant-specific neural variation	**Accuracy:** ~70–75% (subject-independent, closed-set phonological categories) **Comparison:** Outperforms subject-specific models on unseen participants *Interspeech 2019 - early cross-participant EEG imagined speech classification*
Agarwal & Kumar (2022)	EEG	**13**	**Cross-participant**	• Subject-independent LSTM trained across 13 participants • Cross-subject evaluation using leave-one-out protocol • Achieves subject-independence without per-participant calibration	**Accuracy:** 73.56% (5-class imagined speech, subject-independent) **RT:** 0.14 s prediction time - suitable for real-time BCI *ETRI Journal 2022 - n=13, largest imagined speech subject-independent EEG study at time*
Mahapatra & Bhuyan (2023)	EEG	**Multiple**	**Cross-participant (cross-dataset)**	• Deep learning model trained on FEIS dataset, transferred to KaraOne dataset • Cross-dataset transfer learning - participants from different recording setups • Demonstrates generalizability across different EEG equipment and protocols	**Accuracy:** 89.01% (FEIS dataset); 82.35% (KaraOne dataset - cross-dataset transfer) **Transfer:** Model trained on FEIS maintains >82% on entirely different KaraOne participants *Journal of Physics Communications 2023*
Wang et al. (2023) - BrainBERT	iEEG	**Multiple**	**Cross-participant**	• BERT-style masked self-supervised pre-training on large unlabeled intracranial EEG corpora • Cross-participant downstream transfer via shared pre-trained iEEG representations • Foundation model architecture - single pre-trained model fine-tuned for multiple tasks and participants	**Evidence:** Downstream decoding accuracy improvements over supervised baselines across multiple participants **Architecture:** First iEEG SSL foundation model - enables cross-participant transfer without full retraining *ICLR 2023 - first neural SSL foundation model for intracranial recordings*
Yuan & Makin (2024)	ECoG	**Multiple**	**Cross-participant**	• Self-supervised pre-training on unlabeled ECoG recordings across multiple participants • Pre-trained representations transferred to speech decoding tasks • Reduces WER compared to supervised-only baselines	**WER reduction:** SSL pre-training reduces WER on ECoG speech decoding tasks vs supervised baselines **Cross-participant:** Pre-trained representations generalize across ECoG participants *arXiv 2024 preprint - verify final journal publication status*
Peterson et al. (2021)	EEG / ECoG	**Multiple**	**Cross-participant + Cross-modal**	• Generalized neural decoder trained across both EEG and ECoG participants simultaneously • CNN + transfer learning enables cross-participant AND cross-modality transfer • Single model architecture handles multiple recording modalities	**Cross-participant:** Decoder generalizes across participants within and across modalities **Cross-modal:** ECoG-trained model transfers to EEG participants and vice versa *Journal of Neural Engineering 2021*
Caucheteux et al. (2022)	MEG / fMRI	**Large cohort**	**Cross-participant**	• Brain-LLM alignment across large MEG/fMRI cohort • Transformer encoding models predict neural responses across participants • Cross-participant brain-language model alignment via shared semantic representations	**R^2 scores:** Alignment between brain and LLM representations reported across participants **Evidence:** Convergence between neural speech representations and deep LM layers generalizes cross-participant *Communications Biology 2022*
Goldstein et al. (2022)	ECoG / iEEG	**Large cohort**	**Cross-participant**	• Transformer encoding models predict ECoG responses during natural conversation across participants	**R^2 alignment:** Cross-participant alignment between LM representations and ECoG activity **Paradigm:** Naturalistic conversational speech - first such analysis at scale

				• Shared computational principles between deep LMs and brain activity generalize cross-participant • Naturalistic speech paradigm - not controlled stimuli	*Nature Neuroscience 2022*
SECTION 2 - CROSS-SESSION GENERALIZATION (2 studies) Longitudinal stability - critical for sustained clinical deployment					
Card et al. (2024)	Intracortical	**1**	**Cross-session (longitudinal)**	• Rapidly calibrating speech neuroprosthesis - 30-minute daily recalibration protocol • RNN decoder with online fine-tuning to neural signal drift • Consistent vocabulary and task structure maintained across sessions over 8+ months	**WER stability:** 2.5% WER sustained across 8+ months of recording sessions **Accuracy:** 97.5% word accuracy maintained longitudinally without full retraining **Calibration:** 30-minute daily calibration sufficient for high performance *NEJM 2024 - demonstrates clinical viability of long-term intracortical BCI*
Degenhart et al. (2020)	Intracortical	**2**	**Cross-session (manifold alignment)**	• Low-dimensional neural manifold alignment to compensate for day-to-day neural signal drift • Stabilizes decoder performance across sessions without re-collecting labeled data • Linear alignment in low-dimensional latent space - computationally efficient • No behavioral data required for alignment - fully unsupervised cross-session adaptation	**Performance retention:** ~95% of single-session decoding performance maintained across sessions **Comparison:** Without alignment, performance degrades substantially across sessions **Method:** Unsupervised manifold alignment - no labeled data required *Nature Biomedical Engineering 2020 - foundational cross-session alignment study*
SECTION 3 - CROSS-LANGUAGE GENERALIZATION (2 studies) Rare - significant gap in the field					
d'Ascoli et al. (2025)	EEG / MEG	**723**	**Cross-language** (+ cross-participant)	• Single CNN+Transformer architecture applied across English, French, and Dutch speech datasets • No language-specific architectural modifications - same model for all 3 languages • Performance scales with training data size across all three languages independently	**Languages:** English, French, Dutch - 3 typologically distinct languages **Architecture:** Single model generalizes across all three language datasets **Scaling:** Performance improves consistently with more data per language *Nature Communications 2025 - only systematic cross-language non-invasive decoding study in this review*
Zhu et al. (2021)	HD-sEMG	**Multiple**	**Cross-language** (implicit: EN+ZH)	• CNN-based HD-sEMG electrode configuration optimization applied to both English and Chinese digit classification • Single optimized electrode configuration performs across two typologically distinct languages • Tests generalizability of peripheral articulatory representation across language families	**Accuracy:** 87–90% classification accuracy for both English and Chinese digit sets **Cross-language:** Single electrode configuration performs comparably for both languages *Journal of Neural Engineering 2021 - English+Chinese HD-sEMG digit classification*
SECTION 4 - CROSS-MODAL AND CROSS-TASK GENERALIZATION (5 studies) Transfer between speech modes or recording modalities					
de Borman et al. (2024)	ECoG	**16**	**Cross-modal** (speech modes) + Cross-participant	• Linear model trained on one speech mode (overt) transferred to another (imagined or perceived) • Speech event detection transferred across performed, perceived, and imagined speech modalities	**Detection accuracy:** Per-electrode and per-subject detection reported across all 16 participants **Cross-mode:** Transfer from overt→imagined and overt→perceived quantified **Finding:** Different brain regions and spectral bands contribute differentially to each speech mode *Communications Biology 2024*

				• Cross-participant transfer evaluated simultaneously with cross-mode transfer • Frequency band contributions differ across speech modes - informs transfer strategy	
Tang & Huth (2025)	Multi-modal	**Multiple**	**Cross-modal** (recording modality) + Cross-participant	• SSL + Transformer architecture applied simultaneously across EEG, MEG, and fMRI • Single model decodes semantic language representations across three non-invasive modalities • Cross-participant cross-modal semantic decoding via shared latent representation • Each modality provides complementary spatial/temporal coverage of speech language networks	**Cross-modal:** Semantic accuracy demonstrated across EEG, MEG, and fMRI simultaneously **Cross-participant:** Cross-participant generalization demonstrated alongside cross-modal *Current Biology 2025*
Martin et al. (2025)	ECoG	**Multiple**	**Cross-modal** (electrode config) + Cross-participant	• CNN-based speech mode classifier transfers across electrode configurations and participant anatomies • Cross-electrode transfer using domain adaptation • Cross-participant transfer evaluated across multiple ECoG electrode placements	**Transfer:** Classification accuracy maintained across electrode configurations **Cross-participant:** Transfer across participant anatomical variability demonstrated *Journal of Neural Engineering 2025 - early access*
Alonso-Vazquez et al. (2025)	EEG	Not reported	**Cross-modal (task: overt→imagined)**	• Multi-condition EEG data used to improve imagined speech decoding via pronounced speech transfer • Pronounced (overt) speech EEG data augments imagined speech decoding via domain adaptation • Transfer learning from data-rich overt condition to data-scarce imagined condition	**Transfer:** Accuracy improvement for imagined speech decoding when using multi-condition data **Method:** Overt→imagined speech transfer via domain adaptation and data augmentation *Frontiers in Neuroinformatics 2025*

The strongest cross-participant evidence in this review comes from large-scale non-invasive studies where multi-participant training is the default assumption rather than a bonus result. Défossez and colleagues trained a contrastive self-supervised model across 175 participants simultaneously, achieving 41% top 1 accuracy on held-out participants -zero-shot, with no individual calibration [44]. d'Ascoli and colleagues extended this to 723 participants across seven datasets and demonstrated explicitly that cross-participant performance scales with the number of training participants [46]. For invasive modalities, where participant counts are smaller, Singh and colleagues pooled 25 sEEG recordings to show that a group decoder outperforms individual models -the most direct demonstration that shared latent neural representations exist across participants in invasive speech BCIs [38]. Chen and colleagues achieved PCC=0.804 for cross-subject acoustic reconstruction across 48 ECoG participants [30], establishing that large ECoG cohort training is feasible even without the subject count accessible to non-invasive studies.

Transfer learning and self-supervised pre-training are the most consistently effective methods for achieving cross-participant generalization. The BENDR pre-training framework [10], BrainBERT for intracranial signals [62], and Yuan and Makin's ECoG SSL pre-training [64] all demonstrate that pre-trained shared representations improve downstream per-participant performance -reducing the amount of new-participant data required without eliminating the need for any fine-tuning entirely. The critical remaining question is how much individual fine-tuning will always be necessary given the known variability in cortical anatomy, electrode placement, and individual speech neurology. No study in this review achieves zero-shot cross-participant decoding for invasive modalities.

6.3 Cross-Session, Cross-Language, and Cross-Modal Evidence

Cross-session generalization -the ability of a decoder to maintain performance across recording sessions days or weeks apart without full retraining -is the most clinically urgent form of generalization and the most poorly evidenced in this review. Only two studies address it directly. Card and colleagues demonstrated 97.5%-word accuracy sustained across eight months of daily use in a single ALS participant via a 30-minute daily calibration protocol [2]. Degenhart and colleagues showed that low-dimensional manifold alignment preserves approximately 95% of single-session decoding performance across sessions without collecting any new labeled data [33]. Both results are clinically meaningful, but both involve only two participants across two laboratories. A field whose clinical goal is daily, long-term, reliable communication has produced exactly two cross-session generalization studies. That gap is stark.

Cross-language generalization is equally sparse. d'Ascoli and colleagues stand alone as the only study to systematically evaluate a single decoder across multiple languages -English, French, and Dutch -at population scale [46]. Zhu and colleagues demonstrated comparable HD-sEMG performance for English and Chinese digit classification using the same electrode configuration [54], but only for digit sets. The remaining 94% of included studies are English-only. For a technology intended to serve communicatively impaired individuals worldwide - including the hundreds of millions for whom English is not a first language -the language coverage of the existing evidence base represents a profound access gap that the field has not yet begun to address systematically.

Cross-modal generalization -transferring a decoder trained on one recording modality to another -appears in five included studies and represents a nascent but important direction. Peterson and colleagues demonstrated that a single CNN decoder trained jointly on EEG and ECoG data transfers across modalities [66]. Tang and Huth showed cross-participant cross-modal semantic decoding simultaneously across EEG, MEG, and fMRI [50]. De Borman and colleagues quantified the performance cost of transferring an ECoG decoder across speech production modes -overt, imagined, and perceived -in 16 participants [67]. These studies collectively suggest that shared neural representations for language exist across recording modalities and speech conditions, but that transfer is not free: there is always a performance cost when the recording context changes.

6.4 What Generalization Actually Requires

Taken together, the evidence in this section points to three conditions that consistently enable generalization: sufficient participant scale for pre-training; architectural choices that learn modality-level rather than individual-level representations (SSL, contrastive learning, foundation models); and evaluation designs that treat generalization as a primary outcome rather than a secondary observation. The field's dominant single-participant, within-session study design systematically prevents any of these conditions from being met. Shifting towards multi-participant training consortia, standardized evaluation protocols that include held-out participant testing, and explicit cross-session evaluation windows would produce a literature far better equipped to answer the question that matters most to patients and clinicians: will this work for someone other than the person it was built for?

7. Discussion

7.1 What the Evidence Actually Shows

The story this review tells is one of genuine, extraordinary progress - and one of deeply uneven foundations. Over the past decade, brain-to-speech decoding has moved from theoretical possibility to clinical demonstration: real people who have lost the ability to speak are communicating through neural signals at rates approaching conversational speed, with error rates that rival noisy telephone calls. The arc from Herff and colleagues in 2015 to Card and colleagues in 2024 - from 25% WER with classical models to 2.5% WER with RNN-LM systems,

sustained over eight months [2], [6] - is a legitimate landmark in biomedical engineering. These results are not simulations. They are not in healthy participants but they are in people who needed them.

But the foundations beneath that headline number are narrow. The best-performing systems exist for one or two participants per study, use fixed closed vocabularies that do not reflect the open-ended nature of real communication, and have been evaluated by the same laboratory that developed them. The 56% of included studies showing no cross-participant or cross-session generalization is not a failure of individual researchers - it is the predictable consequence of a field structured around small, heroic demonstrations rather than scalable, reproducible evidence. Both things are true simultaneously: the progress is real, and the evidence base for clinical scalability is thin.

7.2 Three Priority Research Directions

The evidence synthesized in this review points clearly to three areas where targeted investment would yield the greatest near-term return for clinical translation.

The first is standardized evaluation protocols. The benchmark comparability crisis documented in Table 5 and Section 5.5 is not an inevitable feature of a diverse research field - it is an addressable problem. The speech recognition community solved the analogous problem decades ago with standardized benchmarks (LibriSpeech, TED-LIUM, CHiME) that allow every new system to be compared against the same baseline on the same data [68]. The brain-to-speech community lacks any equivalent. The practical consequence is that every new paper creates its own evaluation context, making it impossible to determine whether a new architecture genuinely advances the field or merely exploits a more favorable experimental setup. Standardized open-vocabulary benchmarks with pre-specified held-out participant evaluation - analogous to the Brain-to-Text Benchmark 2024 initiative [69] - are a necessary precondition for the field to measure its own progress honestly.

The second priority is multi-participant training infrastructure. The evidence from this review is unambiguous: cross-participant generalization scales with the number of training participants [46], self-supervised pre-training on large unlabeled neural corpora improves downstream per-participant performance [10], [64], and group decoders trained across participants outperform individual models [38]. Yet the structural incentives of the field push in the opposite direction - towards novel single-participant experiments that produce publishable results faster than consortium-level data collection. Shifting this requires coordinated data sharing initiatives, standardized recording protocols that enable cross-institution pooling, and grant structures that reward multi-site replication alongside first-in-field demonstrations. The open dataset releases of Nieto and colleagues [35], Berezutskaya and colleagues [70], and Gwilliams and colleagues [71] represent exactly the right direction - but they remain exceptions rather than the norm.

The third priority is language model integration at scale. Only 14 of 71 included studies (20%) integrate a language model, despite the evidence that LM integration is the single largest discrete driver of WER reduction in open-vocabulary systems [2]. The barrier is not algorithmic - transformer-based LMs are freely available and straightforward to integrate post-hoc. The barrier is data: brain-language paired training sets large enough to support meaningful LM alignment remain scarce, particularly for non-English languages and clinical populations. Investing in the collection, annotation, and open sharing of neural-language parallel corpora - analogous to what Common Voice [72] has done for speech recognition - would unlock LM integration for the 80% of the literature currently operating without it.

7.3 The Clinical Translation Gap

Brain-to-speech BCIs have been described as being on the cusp of clinical translation for most of the past decade. The evidence from this review suggests that description requires qualification. For intracortical systems, the clinical performance evidence is compelling - sustained WER below 10%, communication rates approaching natural

impaired speech, and at least one system operating reliably over an eight-month horizon [2]. The pathway from research to regulated medical device exists and is being actively pursued [73]. But the clinical translation of ECoG and non-invasive systems faces a different and more fundamental challenge: the performance-invasiveness trade-off has not yet been resolved at the level needed for widespread deployment.

For patients who cannot or will not accept neurosurgical implantation - which is the majority of the affected population - the best available non-invasive systems still produce WER well above the threshold for comfortable conversational use. The 35-45% WER range of current EEG open-vocabulary systems is functional for pre-specified command sets but falls short of the open-ended communication that ALS and LIS patients require for genuine social participation. Bridging this gap will require not just algorithmic advances but hardware innovation - higher-density EEG arrays, wearable MEG systems based on optically pumped magnetometers [74], and hybrid modality designs that combine accessible recording with richer signal content.

7.4 Foundation Models: Promising but Not Yet Demonstrated

The emergence of foundation model approaches - SSL pre-training on large multi-participant neural corpora, followed by few-shot fine-tuning for new participants and tasks - represents the most theoretically compelling direction for resolving both the generalization gap and the data efficiency problem simultaneously. The scaling law evidence from d'Ascoli and colleagues [46] is directly analogous to the scaling laws that transformed NLP [75]: if neural decoding performance scales with training data volume the same way language model performance scales with text, then the path to generalizable brain-to-speech systems is, in principle, clear. Collect more multi-participant data. Pre-train larger models. Fine-tune efficiently.

The gap between this theoretical clarity and current empirical reality is, however, substantial. Foundation model approaches account for only 2 of the 71 included studies and have not yet been evaluated in clinical populations or on open-vocabulary tasks in the way that intracortical RNN systems have been. The promise is real; the evidence for it in brain-to-speech decoding specifically is nascent. This review endorses the direction but cautions against treating the NLP scaling analogy as proven - neural signals are not text, the number of participants that can ever be recruited for any single study is limited by neurosurgical access, and the idiosyncratic variability of individual cortical topography may impose generalization limits that text data does not.

7.5 Limitations of This Review

This review has several limitations that should be considered when interpreting its findings. The 2015 lower date boundary, whilst principled, excludes foundational pre-deep-learning work that shaped the conceptual framework of the field. The search cutoff of April 2026 means papers published or posted after that date - including potentially significant developments in the rapidly moving LLM-integration space - are not captured. The quality assessment tool was developed specifically for this review and has not been externally validated. As a narrative synthesis, this review does not produce pooled effect size estimates; the heterogeneity analyses in Table 5 identify why pooling is not valid, but they do not replace the quantitative precision that meta-analysis would provide if the evidence permitted it. Finally, the predominance of English-language studies in the included literature means this review's conclusions about decoding performance are themselves English-centric - a limitation that mirrors the field's own.

8. Conclusion

This systematic review of 71 empirical studies published between 2015 and 2026 has synthesized the AI methods, performance benchmarks, and generalization evidence underpinning brain-to-speech decoding research across five neural recording modality groups. The answers to the three research questions are now grounded in evidence rather than extrapolation.

On RQ1 - AI architecture evolution - the field has progressed through eight distinct architecture classes in a decade, from HMM-based speech recognition through convolutional and recurrent networks, encoder-decoder frameworks,

transformers, and self-supervised pre-training to nascent LLM-integrated and foundation model approaches. The architecture that currently delivers the best clinical results (RNN combined with language model rescoring) is not the most recent - reflecting a recurring tension in applied AI between theoretical advancement and the data and deployment constraints of clinical settings. Only 14 of 71 studies (20%) integrate a language model despite this being the single largest driver of WER reduction in open-vocabulary systems.

On RQ2 - performance benchmarks - the field has achieved extraordinary results in optimal conditions: WER as low as 2.5% [2], communication rates up to 78 words per minute [4], and 100% semantic story identification from non-invasive brain signals [48]. These are genuine scientific achievements. They are also, as Table 5 documents across eight comparability dimensions, not directly comparable to one another or to the majority of results in the literature - because vocabulary size, task type, participant population, evaluation methodology, and metric choice vary so substantially across studies that no single performance number can meaningfully represent the state of the field. Any synthesis that implies otherwise overstates the evidence.

On RQ3 - generalization - the finding is stark. Forty of 71 included studies (56%) demonstrate no generalization beyond a single participant and a single recording session. Cross-session evidence exists in exactly two studies. Cross-language evidence in exactly two. A technology whose clinical promise rests on deployment across diverse patients, across daily sessions, across languages other than English, has not yet produced a body of evidence adequate to that promise. The gap is real, it is quantifiable, and it is the field's most important unsolved problem.

None of this diminishes what has been accomplished. People who could not speak are speaking - through intracortical arrays, through ECoG grids, through the extraordinary convergence of neuroscience, engineering, and AI that defines this field at its best. The task for the next decade is to ensure that what works for one person in one laboratory can work for many people in many contexts. That will require standardized evaluation benchmarks, coordinated multi-participant data infrastructure, and a research culture that rewards replication and generalization as highly as it currently rewards novelty. The scientific foundations are strong enough. What is needed now is the will to build on them at scale.

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